CHAPTER

19

Composite Restorations

Tooth-colored restorations have always been in demand. The earlier used acrylics did not last long because of inherent drawback of higher coefficient of thermal expansion and higher polymerization shrinkage. To improve upon these drawbacks, filler particles were added to the acrylic resin matrix.

The course of inventions and developments led to the introduction of composites by RL Bowen in 1962. Composite is basically a botanical term, where the clusters of flowers are clubbed giving shape to a different flower. Metallurgically, composite is a combination of two or more materials, having chemically distinct interface between them.

The accepted definition is of Anusavice, which says, "*Composite material is a compound of two or more distinctly different materials with properties that are superior to or intermediate to those of the individual constituent*".

Over the years, composite as a material has seen tremendous improvement; modifying the basic constituents and also the restorative modalities.

COMPOSITION

The components of composite are:

a. Resin matrix
b. Fillers
c. Coupling agents
d. Initiators and accelerators
e. Coloring agents

a. Resin Matrix

The initial resin matrix was bisphenol glycidyl methacrylate (BisGMA) and urethane dimethacrylate (UDMA).

The BisGMA/UDMA was viscous and blending of filler particles was difficult. To overcome problem of viscosity, other matrices were tried having lower viscosity, viz: TEGDMA (Triethylene glycol dimethacrylate). BisGMA and TEGDMA in the ratio of 3:1 is preferred; increase in TEGDMA substantially increases the polymerization shrinkage.

The polymerization shrinkage has been of concern during composite restorations. As the material polymerizes, it shrinks and no longer remains attached to the cavity walls. This further creates stresses at the tooth–restoration interface and open the gaps; subsequently leading to debonding, cracks, secondary caries, staining and postoperative sensitivity.

Polymerization shrinkage leads to:
- Stresses at tooth–restoration interface
- Debonding (loss of attachment)
- Opening of gaps
- Microcracks
- Staining
- Secondary caries
- Postoperative sensitivity

Expanding matrix composites are introduced with the idea to compensate for the polymerization shrinkage. Spiro-orthocarbonates are the expanding matrix used. It is preferably used with epoxy resin, because when used with BisGMA/TEGDMA, it could not produce the required expansion. *A newer matrix system containing a diepoxide monomer and polyol has substantially reduced the polymerization shrinkage*.

Hydrophobic monomers are also used as matrix. Recently, thiolene methacrylate, a low-shrink resin matrix, is being tried. It leads to delay in conversion and gelation. As the gelation is delayed, the impact of stresses is reduced; subsequently, gap formation and other deleterious effects are minimized.

A new hydrolytically stable, ether-based monomer (triethylene glycol divinylbenzyl ether: TEG-DVBE and urethane dimethacrylate: UDMA) has shown substantial improvement in mechanical properties in composites. These composites generated much less stress than conventional resin composites.

b. Fillers

Various types of fillers, with varying sizes, have been used in different composites. Depending upon the size of filler particles, the composites are categorized as follows:

- Microfill (40–50 nm)
- Medium fill (40 nm + 1.0–10 μm)
- Hybrid (40 nm + 10–50 μm)
- Microhybrid (40 nm + 0.6–1.0 μm)
- Nanohybrid (50–100 nm + 0.6–1.0 μm)
- Nanofiller (5–50 nm)

The routinely used fillers are:

i. *Quartz:* Quartz were used in early composites, which were difficult to polish and even abraded the opposing tooth structure.
ii. *Silica:* Pure silica, fused silica and colloidal silica (pyrogenic and precipitated forms) have been used. Glasses such as aluminosilicates and borosilicates are also used.
iii. Tricalcium phosphate and zirconium dioxide
iv. Fillers containing fluoride, such as *Yttrium trifluoride* and *Ytterbium trifluoride*

The size of filler particles has continuously decreased from the traditional to nano-composite materials. (Inorganic phase is nano-sized.) Nanofillers are usually invisible and offer improvement in optical properties. Additionally, nanofillers are capable of increasing the overall filler load due to small particle sizes.

Composites reinforced with 2.0–5.0% ceramic nanofillers (zirconia–silica or zirconia–yttria–silica) have significantly improved mechanical properties of composites, especially the fracture toughness. However, higher percentage of these fillers could not show better results.

Fillers provide:
- Strength
- Rigidity
- Hardness
- Increase in modulus of elasticity
- Decrease in coefficient of thermal expansion
- Decrease in contraction

Mostly, present day composites contain combination of macro- (barium glass, particle size 0.7–2.0 μm) and micro- (pyrogenic silica, particle size 0.04–0.07 μm) fillers.

The advantages of using these fillers are as follows:

- Better polishing
- Kindness to antagonist tooth
- Good esthetic and chameleon effect

Recently, wrinkled mesoporous silica (average particle size 496 nm) has been added along with smaller silica particles to achieve superior mechanical properties.

c. Coupling Agents

The coupling agents are used to bind filler particles to the resin matrix. Apart from binding, these agents allow flexible polymer matrix to transfer stresses to stiffer filler particles. The coupling agents also provide hydrolytic stability by preventing the water from penetrating along the filler–resin interface.

Organic silanes, such as γ-methacryloxypropyl-trimethoxysilane, are commonly used as coupling agents. (Silane molecule undergoes hydrolysis and forms a silanol group, which forms a bond with filler particles and carboxyl group; subsequently, facilitates polymerization with the resin matrix.)

Presently, 10-methacryloxydecyl-trimethoxysilane (more hydrophobic and better suited as coupling agent) is being used.

Organosilane acryltriethoxysilane (ATES) is used in titanium oxide reinforced nanocomposite; increases the dispersion and linkage of titanium oxide nano-particles within the resin matrix.

Recently, the process of sol-gel surface modification of filler (quartz) particles, substantially improved bonding of fillers with the resin matrix.

d. Initiator and Accelerator

Benzoyl peroxide is used as initiator in chemically cure composites. (Earlier composites were chemically cure.) An accelerator, a small amount of tertiary amines, such as dimethyl-p-toluidine, is added in the monomer. The polymerization is initiated by the release of free radicals from the reaction between benzoyl peroxide and the amine.

Since amines were mostly unstable, a stable accelerator, p-toluidine sulfinic acid, showed effective results.

Light cure composites, which utilizes benzoin methyl ether (UV light activated) and camphorquinone (visible light activated) as initiator.

Dimethylamino ethyl methacrylate is used as accelerator. The differences between chemically-cure and light-cure composites are presented in Table 19.1.

The differences between the two lights used for curing composite are summarized in Table 19.2.

e. Coloring Agents

Aluminum oxide and titanium dioxide in 0.001–0.007% by wt. are commonly used as coloring agents.

It has been established that organic matrix influences the translucency of the composite material. (Translucency

Table 19.1: Difference between chemically-cured and light-cured composites

Chemically cured	*Light cured*
Polymerization is central	Polymerization is peripheral
Curing is in one phase	Curing is in increments
Sets within 45 seconds	Sets only after light activation
No time for manipulation	Plenty of time for manipulation
More wastage Not properly finished	Less wastage Better finish and polish

Table 19.2: Difference between UV light and visible light used for curing composites

UV light	*Visible light*
Works at 360–400 nm	Light range is 400–500 nm
Intensity falls with time	Intensity remains the same
Injurious to operator and patient eyes	Not injurious
Greater depth cannot be cured	Greater depth can be cured

depends upon absorption and scattering.) The organic component absorbs light; however, weak in scattering (Scattering is due to refractive index mismatch between filler particles and the matrix.) The scattering of light in composite would increase by increasing the mismatch in refractive index between filler particles and the matrix.

EVOLUTION OF COMPOSITES

The evolution of composites vis-à-vis composition, filler particles, their size and percentage, etc. is elaborated in Table 19.3.

PROPERTIES OF COMPOSITES

Physical properties vary between the three basic types of composites, i.e. conventional/macrofilled, microfilled and hybrid (Table 19.4).

ADVANCES IN COMPOSITES

i. Flowable Composites

The flowable composites are characterized by substantially reducing the filler contents, so as to decrease the viscosity of composite material (Fig. 19.1). They serve as liner to absorb the shrinkage/contraction of the overlying composite restoration.

The characteristic features are:

- Filler content is 20–25% less than that of the traditional hybrid composites.
- Because of lesser amount of fillers loading, the flow is increased.
- Depth of cure is appropriate (approximately 6.0 mm).
- Stickiness to the instrument, which makes it difficult to smoothen the material.
- Mechanical properties, like compressive strength, tensile strength, flexure strength and toughness values, are less as compared to conventional composites.



Fig. 19.1: Flowable composites

Table 19.3: Evolution of composites vis-à-vis composition, filler particles, size and percentage of filler

Year	*Nomenclature*	*Resin matrix*	*Filler*	*Percentage (by wt.)*	*Size of filler (μm)*
1962	Conventional Type I	Polymethyl methacrylate (PMMA)	Quartz	30–70%	25–30
1968	Conventional Type II	BisGMA (bisphenol glycidyl methacrylate replaces PMMA)	Quartz (barium added in traces)	30–70%	7–10
1975	Microfilled	BisGMA	Silica replaces quartz	30–70%	0.05
1978	Organic	BisGMA	Quartz	10%	20–30
1989	Hybrid	UGDMA/TEGDMA	Silica	50–70%	20–30 and 0.05
1996	Flowable	TEGDMA	Silica	60%	0.05
1996	Packable	Urethane modified BisGMA	Fumed silica and barium fluoroaluminoborosilicate	82%	0.8
2000	Modified Hybrid	BisGMA and Propanal	Barium glass Pyrogenic silica	80%	0.07–0.7

Table 19.4: Physical properties of Conventional, Microfilled and Hybrid composites

Property	*Conventional*	*Microfilled*	*Hybrid*
Compressive strength (psi)	36000–43000	50,000	60,000
Tensile strength (psi)	7000	7500	10,000 to 13,000
Coefficient of thermal expansion	25×10^{-6}/°C	50×10^{-6}/°C	$30–40 \times 10^{-6}$/°C
Water sorption	0.5%	1.4%	0.5%
Polymerization shrinkage	1.2–1.3%	1.5–2.0%	1.0–1.5%
KNH	50–60	20–30	50–60

Uses

- Preventive resin restorations (minimal Class I)
- Abfraction lesions (cervical small lesions)
- Filling material in low stress areas
- Pit and fissure sealant
- As liner in proximal box of Class II preparation
- Repairing porcelain/silver amalgam restoration
- Rebuilding worn contact areas in composite restorations
- Tunnel restorations
- Core build-up
- Cementing agents for porcelain restorations
- Sealing gingival line/point angles in Class II restoration
- Bonding of orthodontic brackets
- Blocking small undercuts in indirect cavity preparation
- Repair of crown margins
- Splinting periodontally involved teeth

Limitations

- High curing shrinkage
- Reduced compressive strength
- Low elastic modulus
- Decreased wear resistance
- Water sorption
- Low radiopacity

Flowable composites are generally contraindicated for Class I and Class II restorations, because of the relatively high stresses in these areas.

Self-adhering flowable composite (Vertise Flow) incorporates glycerophosphate dimethacrylate (GPDM) monomer, which forms chemical bond with the calcium ions of the tooth; successfully tried in low stress-bearing areas.

Flowable composites have been reinforced with titanium oxide nanotubes (n-TiO_2) improving mechanical properties of the composites, with minimum decrease in flowability and radiopacity.

ii. Packable/Bulk-fill Composites

Packable/bulk-fill composites are based on the concept, called PRIMM (polymer rigid inorganic matrix material). The filler/inorganic phase, instead of being incorporated as ground particles, is present as a continuous network/scaffold of ceramic fibers. The fibers are composed of alumina and silicon dioxide. The diameter of the individual ceramic fiber is less than 2.0 μm.

Examples of condensable composites include Solitaire (Heraeus-Kulzer), Alert (Jeneric/Pentron), Surefil (Caulk, Dentsply), Filtek P60 (3M), etc.

The consistency of PRIMM-based composites is similar to that of a freshly triturated mass of silver amalgam. The composite is inserted into the prepared cavity by ejecting from a carrier whose nozzle is preferably made from/coated with wear-resistant teflon polymer (Fig. 19.2A, B). The use of a conventional amalgam carrier is not advised as the hard alumina fibers can scratch/damage the nozzle easily. Each ejected increment is then condensed and light cured.

Most bulk-fill composites are placed in 4.0 mm increment/one bulk and can be cured to that depth with improved curing lights and techniques. Many studies comparing increment versus bulk-fill placement have reported no difference in cuspal deflection or marginal integrity.

Surefil SDR flow (smart dentin replacement), a flowable bulk-fill composite and couple of other

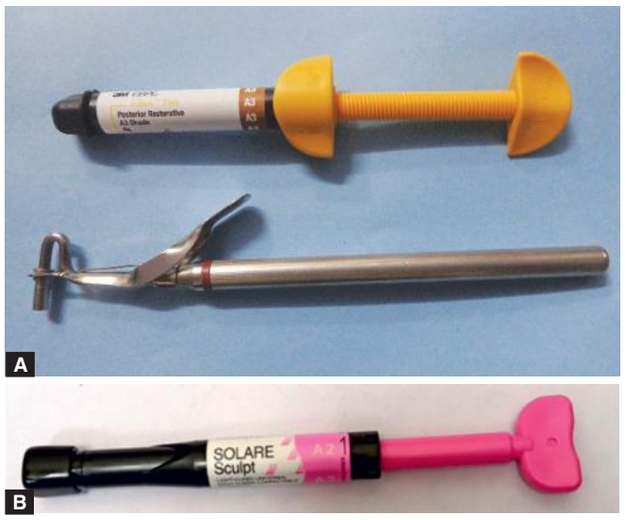


Fig. 19.2: Posterior composite and carrier

commercial preparations (Tetric Evoflow, Hyperfil, etc.), were used as 'base' (dentin replacement) under the composite restorations. The viscosity of the material allow the base to be adequately adapted to cavity walls. Since the filler contents of these composites are less, the base is to be covered by conventional hybrid composites.

Another advancement in bulk-fill composites is the Sonic bulk-fill delivery system, which utilizes sonic handpiece and conventional composites (Sonic fill-Kerr Dental). The sonic energy (vibrations) generated by the handpiece changes the viscosity of the composite, so that it behaves as a flowable material during placement. Once sonic energy is stopped, the composite returns to its conventional form, which is used to fill the upper portion of the cavity. Sonic fill exhibits 1.0–2.0% shrinkage and can be filled to a depth of 5.0 mm, without requiring the separate hybrid composite in the last occlusal increments.

Packable composites present improved properties over conventional ones, viz:

- Increased flexural modulus
- Increased resistance to wear
- Non-stickiness

iii. Antibacterial Composites

To minimize accumulation of plaque on the restorative surface and development of secondary caries, various antibacterial agents have been incorporated in the composite resins.

The agents commonly incorporated are:

i. ***Chlorhexidine:*** Chlorhexidine was tried in an attempt to reduce plaque accumulation; however, that was not successful. The disadvantages of using chlorhexidine are:
   - Population shifts of the micro-organisms
   - Antibacterial activity is short-lived
   - Deterioration of the physical and mechanical properties of the material.
   - Toxic effects of the released material

   Other soluble antibacterial agents added to composite include chitosan and cetylpyridinium chloride. [Benzalkonium chloride (quaternary ammonium compound) could achieve antibacterial effects without changing the physical properties.]

ii. ***Silver:*** The antibacterial effect is attributed to the direct contact of silver with bacteria and not because of release of silver ions. The catalytic action of silver and the hydroxyl radicals results in the structural damage of bacteria. (Phenomenon is referred to as the oligodynamic action.)
   The inclusion of silver into composite does not adversely affect the mechanical properties, like strength, translucency, color stability and depth of cure. Silver does not disrupt the polymerization of composite monomers. Silver-silica glass prepared by the sol-gel method has shown promising antibacterial efficacy.

iii. ***Methacryloxydodecyl pyridinium bromide (MDPB):*** Methacryloxydodecyl pyridinium bromide has been incorporated into the resin composite. The methacryloyl structure of the MDPB molecule co-polymerizes with other methacrylate monomers and is bound to the matrix resin. After curing, there occurs no elution of the antibacterial components from the material. MDPB was found effective against streptococci, but other plaque bacteria remained ineffective. Inclusion of MDPB into composite had no adverse effects on the mechanical properties of the composites.

iv. ***Zinc oxide powder:*** Zinc oxide powder displays antimicrobial properties. Zinc oxide nanoparticles have been found to be more effective than larger particles, both against gram-negative and gram-positive micro-organisms. The antimicrobial properties of zinc oxide nanoparticles are because they generate active oxygen, which inhibits growth of planktonic microbes. Toxicological mechanisms of zinc ions play an important role in biofilm inhibition by inhibiting the active transport and metabolism of sugars. Zinc also act by disrupting enzyme systems of dental biofilms by displacing magnesium ions essential for enzymatic activity of plaque.

v. ***Antibacterial polycations:*** The bacterial cell structure is negatively charged. The polymers retaining positive charge/cationic on their surface act as contact disinfectants. Such polymers include, pyridinium type polymers and polymers activated with polyethyleneimine.
   Quaternary ammonium polyethyleneimine has been reported to be highly effective (antibacterial activity is attributed to the hydrophobic and positively charged long polymeric chains). It is effective against gram-positive and gram-negative bacteria.
   Nanoparticles of polyethyleneimine, even in traces, when added to composite, have shown high antibacterial activity.

vi. ***Newer agents:*** A few agents have been incorporated into resin monomers to have antibacterial activity. Dimethyl-hexadecyl-methacryloxyethylammonium iodide (DHMAI), a quaternary ammonium derivative, when added in 7.5% concentration, provides strong antibacterial effect. The strength and hardness of the composite remain unaffected by its addition. Another such compound,

methacryloxy-ethyl phosphoryl-choline (MPC), did not provide improved antibacterial efficacy and also affect mechanical properties of the resin composite. A few authors have shown protein repellent and antibacterial capabilities of 3.0% MPC. MPC facilitate reduction in biofilm formation and lactic acid production without compromising mechanical properties. A combination of both has shown promising results. Recently, eutectic solvent formed by mixtures of choline chloride and urea (a modified form of benzalkonium chloride) has been successfully incorporated into resin composite. The antibacterial efficacy is maintained for a longer time and the material has shown biocompatibility with the composite.

vii. Another antimicrobial agent used is Halo, which is added up to 1.0% by weight to commercial composite.

iv. Expanding Resin Matrix for Composites

Composite resins that would expand during polymerization are desirable to minimize failure of the restoration due to polymerization shrinkage. Spiro-orthocarbonates (SOCs) have been tried as expanding matrix in composites. It has been established that SOCs were compatible with BisGMA-TEGDMA mixtures. These formulations exhibited less shrinkage than the conventional resins, but shrinkage could not be completely eliminated.

Epoxy resins have shown favorable results. The epoxy resins contract approximately 3.4% and SOCs expand approximately 3.6%; therefore, combining these two will achieve a net polymerization expansion. It is observed that spiro-orthocarbonate in combination with epoxy resin decreases polymerization shrinkage, increases toughness and decreases water permeation.

v. Bioactive Composites

Calcium phosphate and its modified varieties, when used as filler, serve as bioactive liners and bases, enhancing the remineralization process. Amorphous calcium phosphate hybridized with glass forming agents is also used as filler in composites, known as '*smart composite*'.

vi. Ormocer

Ormocer (acronym of organically modified ceramic), is an improved version of composite; can be used for both anterior and posterior teeth.

Chemically, it differs from composites in matrix constituent. Traditional composites are based upon BisGMA, TEGDMA, etc. The ormocer matrix consists of ceramic polysiloxane (silicon-oxygen chains). The polysiloxane is biocompatible and exhibits low shrinkage. Ormocer contains 20–25% matrix and the rest are fillers and coloring agents.

Table 19.5: Physical properties of ormocer

Bending strength (3-point bending test)	100–160 MPa
Volume shrinkage	1.9%
Modulus of elasticity	10–17 GPa
Coefficient of thermal expansion	$17–25 \times 10^{-6}/°C$
Water uptake	<1.2%
Degree of conversion of monomer	>25%

[Ormocer exhibits 1.9% volume shrinkage (lowest in resin-based composites)].

Ormocer is considered to be more biocompatible than the currently available composite resins owing to its crosslinked structure with least amounts of residual monomer. The bonding agent used with ormocers is an ormocer-based agent with functional calcium group for enhancing the bond strength to the tooth structure.

The commercial preparation, Admira Fusion (VOCO) with nano-silicon oxide (SiO) particles exhibited polymerization shrinkage up to 50%. The degree of conversion, being more than 95%, the material is identical to indirect composite; sustain high degree of lustre and is stain resistant.

The physical properties of ormocer are tabulated in Table 19.5.

Advantages

- Biocompatible
- Reduced polymerization shrinkage
- High abrasion resistance, can be used in stress-bearing areas
- Esthetically pleasing, available in different shades
- Anticaries properties due to fluoride release
- Safe handling and easy manipulation
- Cost-effective

vii. Silorane

Silorane, the combination of siloxanes and oxiranes, was introduced in order to provide a hydrophobic composite, since high water sorption limits the long-term physical strength of the conventional composite.

Silorane composite contains three components—initiating system (comprising camphoroquinone), an iodonium salt and an electron donor (ethyl dimethyl-aminobenzoate).

The inorganic fillers are a combination of fine quartz particles and radiopaque yttrium fluoride. The silane

layer is modified with a silanization process. The silane layer increases the hydrophobic character of the surface of the filler and acts as the interface between filler and resin, facilitating the reinforcement of the resin with the filler particles. An important function of the silane layer in the silorane technology is to prevent an attack of the acidic Si–OH group of the quartz, resulting in undesired initiation of the cationic polymerization process.

Composition	
Filler	76% (particle size 0.04–1.7 μm)
Silorane	23%
Initiator	0.9%
Stabilizer	0.13%
Pigments	0.005%

Advantages

- Biocompatible
- Reduced polymerization shrinkage
- High abrasion resistance, can be used in stress-bearing areas
- Esthetically pleasing, available in different shades
- Anticaries properties due to fluoride release
- Safe handling and easy manipulation
- Volume shrinkage is 0.90–0.99 wt%
- Cost-effective

viii. Fiber-reinforced Composite

Fiber-reinforced composites can be divided according to the reinforcement of polymer fillers used. Fiber-reinforced composites have two distinct constituents. The reinforcing component provides strength and stiffness, while surrounding matrix provides workability. Glass fibers are the most commonly used reinforcing fiber in dental applications. Carbon/graphite/metal fibers are also used. Fiber-reinforced composites (FRCs) can also be divided into groups based on length and orientation of fibers; long fibers containing FRCs and short fiber containing FRC. Unidirectional and bidirectional fiber variants are also available.

Fiber reinforced composites have improved mechanical properties and their strength to weight ratio is superior to most alloys used in dentistry.

Glass fibers have documented reinforcing efficiency and good esthetic qualities as compared to carbon fibers. The effectiveness of fiber reinforcement is dependent on many variables, including the type of resin used, quantity of fibers in the resin matrix, fiber length, form, orientation and impregnation with the resin.

50 wt% nanofibers (silica, glass fibers) with size of 3.0 μm has been added to achieve better results.

Advantages

- Non-corrosiveness
- Translucency
- Good bonding properties
- Repair facility
- Facility for both office and laboratory preparations

ix. Nanocomposite

When inorganic phase in composite become nanosized (range of filler particles: 0.1–1.00 μm), they are called nanocomposites (Fig. 19.3).

Nanofillers are extremely small filler particles (usually 0.4–0.8 μm), unable to scatter or absorb visible light (below wavelength of visible light). Nanofillers increase the filler level due to their small particle sizes. However, the increase in nanofillers also increases the surface area, which limits the total amount of filler particles because of their wettability. Increase in filler level results in significant reduction of polymerization shrinkage and dramatically improve the physical properties of nanocomposites.

Filtek Supreme contains nanometric particles (nanomers) and nanoclusters (NCs). Premise is a nanohybrid composed of three different types of filler components: Nonagglomerated 'discrete' silica nanoparticles, pre-polymerized fillers, and barium glass fillers. Ceram-X is an ormocer-based, nanoceramic-composite. Ceram-X contains glass fillers (1.1–1.5 μm) and methacrylate modified silicon-dioxide nanofiller (1.0 μm).



Fig. 19.3: Nano-ceramic composite

CURING OF COMPOSITES

Composite is cured using visible light emanating through a specific light cure unit. The terminology used to describe output of light source from light curing units is:

- *Radiant power (Watt/W):* Energy per unit time (joules/second)
- *Radiant irradiance [Watt/square centimeter* (W/cm^2)*]:* Power emitted from the curing unit
- *Radiant energy (Joule/J):* Energy from the source
- *Radiant exposure [(Joule/square centimeter)* (J/cm^2)*]:* Energy received per unit area

A light curing unit with a minimal light output of 550 lux is considered appropriate for dental use. The amount of light radiation reaching a given point in the material depends on the following factors:

i. *Lamp output intensity:* The energy output from the lamp; governed by lamp power rating.
ii. *Exposure time:* The exposure time is adjusted to maximize energy output.
iii. *Distance from light source to material:* It is established that part of the emitted light is absorbed by air through which it passes; the source of light should be close to the target surface (without touch).
iv. *Curing depth:* Light is gradually absorbed within the material. For the depth of curing, the amount of energy required will be equal to the intensity and duration of exposure.
v. *Presence of elements between light source and target material:* The amount of light absorbed by intervening elements varies according to the thickness and optical behavior of the elements. Enamel is transparent and allows the passage of ample amount of light; however, dentin is considerably less transparent and allow virtually no light to pass through. The passage of light through bands and preformed crown is similarly governed by transparency and thickness.

LIGHT-CURING UNITS

Various modalities of light and light cure units are as follows.

1. ***Ultraviolet light cure units:*** Ultraviolet light cure unit utilizes ultraviolet light in the range of 365 nm; however, these units are no longer used.

Disadvantages

- Prolonged exposure time (90 seconds)
- Harmful effects of ultraviolet radiation to human eyes (corneal burns and cataract formation)
- Poor curing of resin
- Unable to penetrate deeper composites

2. ***Quartz-tungsten-halogen devices:*** Quartz-tungsten-halogen (QTH) devices are the most widely used light-curing units and contain a quartz bulb with a tungsten filament in a halogen environment. The units irradiate both UV and white light. The UV radiations need to be filtered to transmit light in the violet-blue spectrum that matches the photoabsorption range of Camphorquinone. Less than 0.5% of the total light produced in a QTH is suitable for curing; most is converted to heat. The infrared filters are inserted just before the fiberoptic system to filler light. A small fan is employed to dissipate unwanted heat from the filters and reflector. Usually, filters degrade with time due to the heating and cooling cycles. QTH-curing lights work at wavelengths of 400 to 500 nm with output ranging from 400 to 800 mW/cm^2 (Fig. 19.4). The halogen bulb usually last for 50 hours and had to be replaced.

Advantages

- Easy to install
- Relatively inexpensive

Disadvantages

- Slower cure time (about 40 to 60 seconds).
- Units are relatively large and cumbersome.
- Light (bulb) decreases in output with time and need frequent replacement.
- Generate high temperature.
- Require a filter and ventilating fan.
- Darker shades of composite have lesser degree of conversion as compared to light shades when cured at same intensity and for same duration.



Fig. 19.4: Quartz-tungsten-halogen light cure unit

Turbo tips provided with some units offer greater curing intensity at the site of the exit of curing light. The enhanced halogen curing light unit is also available, having special tungsten quartz halogen bulb, whose performance does not degrade with time.

3. ***Light-emitting diode:*** Initially, low-power blue light emitting diodes having a power output of 7 mW per LED were utilized. The modified LEDs, built on gallium nitride technology, had a power output of 3 mW (400-fold increase). These units are cordless, small, light weight, and battery-powered (Fig. 19.5). They do not require filters because they emit light at a specific wavelength within the range of 400 to 500 nm. All the emitted light is useful, resulting in high energy performance of the curing

Advantages

- Constant effectiveness without any drop in intensity with time (the diodes do not require frequent replacement).
- Cooling fan is not needed (no heat generation during curing).
- Depth of curing with LED units is higher than QTH devices.
- LED unit has no bulb or filter that require maintenance.
- Potential lifetime is 10,000 hour.
- Some units have integrated microprocessor to control the light intensity, which remains constant at all times; irrespective of whether the battery is charged or running low.

Disadvantages

- Batteries must be recharged.
- Cost more than conventional halogen lights.
- Curing time is slower than that of plasma-arc curing lights and some enhanced halogen lights.



Fig. 19.5: LED curing light

light. The spectral output falls between 410 and 490 nm.

4. ***Plasma arc curing light:*** Plasma-arc curing lights are high-intensity light-curing units (900 mW/cm^2). They have more intense light sources (fluorescent bulb-containing plasma), allowing for shorter exposure times. Light is obtained from an electrically conductive gas (xenon) called plasma. The light emitted is filtered to minimize transmission of infrared and UV energy and to allow emission of blue light. This also helps remove the heat from the system. Because a high-intensity light is available at lower wavelengths, these units are able to cure composites with photoinitiators other than camphorquinone. An exposure of 10 seconds from a plasma arc light is equivalent to 40 seconds from a QTH light. These systems work at wavelengths between 370 and 450 nm.

Advantages

- Units have a high energy output and short curing time.
- Units show higher conversion rates and depth of cure as compared with QTH units.

Disadvantages

- Heat production must be controlled.
- Lamp (bulb) replacement is costly.
- Most devices are large, heavy, and bulky.
- They have low-energy performance.
- Filters and ventilating fan are required.
- Composites cured with a Plasma-arc unit have shown more polymerization shrinkage than with QTH units.
- Expensive.

5. ***Argon laser curing light:*** Argon Laser produces blue light having the highest intensity, 800 mW/cm^2. These units work within a limited range of wavelengths; do not require filters, require shorter exposure time for curing composites without generating heat. They work at specific bandwidths of light in the ranges of 454 to 466 nm, 472 to 497 nm, and 514 nm. Because laser is a narrow beam of coherent light, no loss of power over distance occurs as seen in QTH units. Argon laser-curing lights are preferred for inaccessible areas. The intensity of laser required for curing is 250 mW ±50 mW.

Advantages

- Curing time is short (5–10 seconds)
- Polymerization is uniform and is not affected by the distance between the material and the light source.
- Uniform depth of cure
- Degree of conversion of materials is higher when cured by lasers as compared to conventional halogen light

Disadvantages

- Curing tip is small; so more time is needed to cure composites
- Narrow spectral outputs
- Units are heavy
- Expensive

6. ***Turbo light guide:*** Conventional light guides provide diameter of light source end and the target end equal in size. In contrast, in turbo light guide, the entrance is made larger than the exit to increase

the irradiance (number of photons provided to the output target area). This arrangement delivers the same power (watts) over a smaller area, increasing the irradiance (mW/cm^2). The optical designs of the turbo light guides provide '*focusing effect*' onto the target material.

Radiometer

The light intensity and output of a light-curing unit can be monitored using a portable or chairside built-in radiometer. A radiometer measures the number of photons per unit area, and the unit time. The radiometer cannot discriminate the light energy of the photoinitiator; limiting the measurement of the real value (measure all light energy) (Fig. 19.6).

Techniques of Curing

The techniques of light curing composites are:

a. ***Soft-start curing:*** In soft-start technique, initially low intensity curing is utilized, followed by a high intensity curing. Various light-curing units provide soft-start exposure sequences (100 mW/cm^2 for 10 seconds, followed by 600 mW/cm^2 for 30 seconds). Soft-start polymerization is divided into three techniques—stepped, ramped, and pulse-delay.

i. *Stepped:* The restoration is initially cured at low intensity to contour and shape the restoration. It is followed by second exposure to completely cure the finished restoration.

ii. *Ramped:* In this method, intensity is gradually increased or 'ramped up.' The intensity is increased with time either by bringing the light closer towards the tooth or using a curing light designed to increase in intensity with time.

iii. *Pulse-delay:* In the pulse-delay method, a series of exposure pulses is used, each separated by a rest period. An initial exposure of up to 1 J/cm^2 is effective in reducing shrinkage stresses. During the rest period, polymerization reaction occurs at a reduced rate.



Fig. 19.6: Curing light meter

b. ***High-intensity curing:*** High-intensity curing allows for shorter exposure times for a given depth of cure. A depth of 2.0 mm can be cured in 10 seconds with a plasma arc light and 5 seconds with an argon laser-curing light, as compared to 40 seconds by a quartz- tungsten-halogen lamp. The relationship between energy density and shrinkage may translate into increased stress levels, but do not result in high degrees of conversion or superior mechanical properties.

Disadvantages

- Short exposure times cause accelerated rates of curing and insufficient time for stress relaxation (lead to greater shrinkage stresses).
- High-intensity light curing has a narrow wavelength range (the wavelength range of the light source must be coincident with the photoinitiator); subsequently, inadequate curing.
- Considerable heat is generated.
- May not produce the same type of polymer network during curing.
- Result in more cytotoxicity than a longer curing time with lower intensity.

c. ***Extraoral curing:*** Extraoral curing is used for the fabrication of indirect composite restorations, processed in the laboratory. The laboratory photocuring units utilize various combinations of light, heat, pressure, and vacuum to increase the degree of polymerization. It is reported that laboratory units, which provide light curing in conjunction with heat and nitrogen pressure, result in a significant increase in hardness and tensile strength of composites.

Clinical Guidelines

The following clinical recommendations improve the use of curing lights, vis-à-vis the success of the restorations.

- Monitor the output of the radiant energy periodically; replace the unit, if necessary.
- The tip should be cleaned regularly.
- Protect the eyes of operator and the assistant.
- Place tip of curing light directly over the surface (as close as possible).
- In proximal restorations, the curing tip is moved buccolingually.
- Avoid overheating; use low power setting, especially in deep cavities and restorations near the gingival tissues (gingiva should also be covered with gauze).
- Tips should be disinfected regularly.

CAVITY PREPARATION

The factors, which influence the choice of composite restoration/factors, which contraindicate composite restoration are:

- *Age of the patient:* In very young age, mostly during rampant caries stage, the composite restorations are avoided.
- *Caries index of the patient:* Composites are not preferred in higher caries index patients.
- *Abnormal occlusal contacts and stresses:* The ceramic filler composites can bear the stresses; however, composites are avoided in heavy occlusal stresses.
- Areas which cannot be isolated properly; composites are not preferred.
- Poor oral hygiene; composite is not the right choice.

General Consideration

The cavity for composite restorations is usually conservative. It is advisable to shape the cavity following the principle of resistance and retention form. Preferably the outline form of the cavity should include the carious fissures/areas. The concept of 'extension for prevention' is not implemented.

The steps followed for cavity preparation are:

a. ***Anesthesia:*** Anesthesia is indicated in almost all the patients, more so, while restoring lesions near the gingival tissues. Anesthesia acts in two ways: (i) Eliminates the apprehensions of the patient, and (ii) help in isolation procedure. Anesthesia also reduces salivation, thereby keeping the operator and the patient at ease. However, prior to anesthesia, history of the patient and the possibility of toxic/allergic reactions must be looked into.

b. ***Prophylaxis:*** Thorough prophylaxis of the oral cavity, especially the area concerned is mandatory for composite restorations. Bleeding gums are one of the causes for failure of such restorations. Polishing with pumice powder and water is preferred since mixing of glycerine and other ingredient can hamper with the acid etch technique, subsequently the success of the restoration.

c. ***Shade selection:*** Color selection should be carried out in such a manner so that the restorations should not look isolated, but should merge with rest of the tooth.

For shade selection, following factors should be kept in mind:

- The selection of shade/color should preferably be carried out in day light. In case, the day light is not available, the light source should be kept at a distance of six feet.
- Patient and the attendant should not be involved in the process; only the operator, or if need be, the operator's helper can carry out the shade selection.
- Ask the patient to rinse or sip water before shade selection (dry tooth appear lighter).
- Shade guide should also be wet during shade selection.
- Prior to shade selection, judge visually and think of the possibility of 2 or 3 shades. These 'possible' shades should be kept aside for final matching. There are plenty of shade guides available. Different manufacturers' have their own shade guide; however, VITA shade guide or VITA 3D is the most accepted one.
- Keep one selected shade on the lingual aspect of the teeth and move buccally (Fig. 19.7A). Examine the shade for 20 seconds only. If more time is needed, the eyes should be rested by looking at a blue or violet object for a few seconds. The change in color at the cervical (darker) and incisal (translucent) can be made accordingly.
- For final checking the shade, place a small amount of selected composite on the adjacent tooth and cure (Fig. 19.7B). In case it matches, the cured composite is removed.
- The final shade is recorded on patient chart (Fig. 19.7C to F).

Shade selection vs shade matching: When the color (shade) of tooth is selected by the operator, it is shade selection; whereas, when the same shade is matched with the composite resin (may be by the operator or the laboratory technician), it is referred to as 'shade matching'.

Isolation

Composite resins are susceptible to moisture; therefore, complete control of moisture during composite restoration is mandatory. Isolation from moisture and also from soft tissues have been discussed in Chapter 11. In brief, the isolation can be accomplished by rubber dam or cotton rolls. The choice of rubber dam or cotton rolls depends upon the anticipated time for the restoration. If longer time period is anticipated, rubber dam is indicated; otherwise, cotton rolls are sufficient for the procedure.

a. ***Rubber dam:*** It is advisable to isolate one tooth mesial and one tooth distal to the one where cavity preparation is planned in posterior teeth. In anterior teeth, preferably all the anterior teeth should be isolated so as to have clear access during lingual/facial approach (Fig. 19.7G). A wedge is usually needed where the proximal lesions are extended subgingivally, to expose the extent of the lesion at the gingival end. Similarly, in Class V cavities, the retainer is pushed more gingivally to expose the gingival extension of the lesion. For placing the wedge or retainer, care should be taken not to injure the rubber dam or the gingiva.

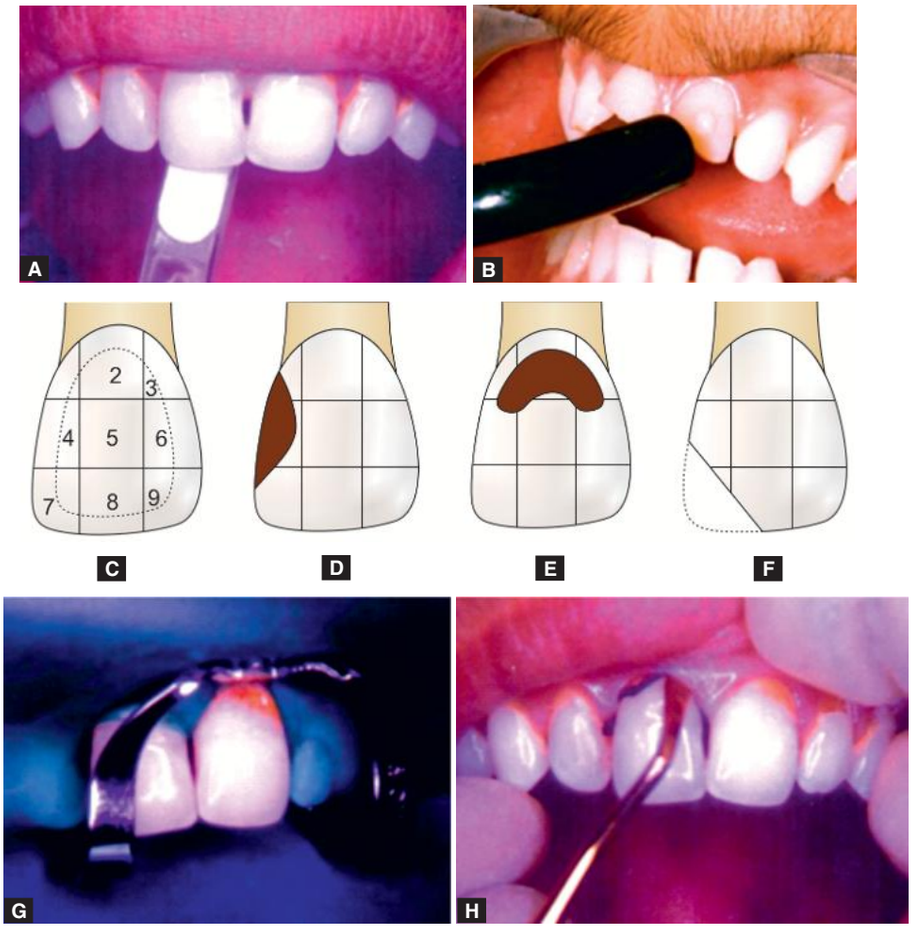


Fig. 19.7: Shade selection and isolation: **(A)** Keeping shade tab on lingual side (shade selection); **(B)** Small amount of selected composite placed on the tooth (shade matching); **(C–F)** Final shade is recorded on patient chart and the menu is prepared keeping in mind the cavity to be restored; **(G)** Rubber dam placement; **(H)** Use of gingival cord

b. ***Cotton rolls:*** Prefabricated cotton rolls or the available cotton drawn into rolls can be used to isolate the area. The moment, the cotton rolls are half soaked in saliva, these are changed and replaced with fresh cotton rolls. A few patients, especially children, are non-cooperative with the placement of cotton rolls.

c. ***Retraction cords:*** Retraction cords are employed to facilitate better accessibility and to control gingival bleeding and the crevicular fluid seepage (Fig. 19.7H). These cords, mainly impregnated in astringents, help in receding the gingival tissue temporarily. Retraction cords are available in different shapes and sizes to suit the various clinical situations. The cord is gently placed in the sulcular area with the help of plastic instruments. Use of probes should be avoided as these may injure the gingiva leading to bleeding from the area.

The time for cord placement is dependent upon the chemical used and the retraction required. Mostly these are instructed on the commercial preparations. The cord should not be pressed unnecessarily and should not be kept longer than required. This can lead to ischemia of the gingiva.

The cord should be removed gently without injuring the gingival tissue and without leaving a loose strand. Loose strands will lead to defective restorations.

Importance of C-Factor

C-factor is the ratio of bonded to the unbonded surface area in a tooth preparation. It is an important clinical consideration regarding its effects on polymerization shrinkage. Throughout the entire polymerization process, plastic deformation or flow of the composite occurs along the unbonded surface that might partially relieve the induced shrinkage stress. Such compensation through flow is affected by the C-factor of the restoration. An increase in the number of bonded surfaces results in a higher C-factor and greater contraction stress on the adhesive bond.

Types of Cavity Preparation

The cavity preparation for composite is of two types:

a. ***Conventional cavity preparation:*** The conventional cavity preparation follows Black's principles. The outline form includes extension of susceptible lesions, proper uniform depth and the external walls in accordance with enamel rods.

   Bevels are usually recommended on labial surfaces of anterior teeth so as to merge the color of the composite and the enamel. Lingual bevels can be avoided, since that might lead to stresses; also the color merging is not important in that area. In Class V cavities, bevels are given only on occlusal walls and the gingival walls are spared. In posterior restorations, bevels are not indicated, since the thin layers of composites might chip off under stresses leading to marginal gaps.

b. ***Modified cavity preparation:*** Modified cavity preparation does not follow Black's principles. The retention part is totally dependent upon the etched enamel and bonding. The carious lesions are excavated and restored separately. This type of cavity preparation conserves more tooth structure and maintain the strength of the tooth. No definite walls and line angles are visible and usually appear saucer shaped. There can be more than one cavity in a single tooth.

Class I Cavity

The conventional Class I cavity is prepared following Black's principles (Fig. 19.8). The depth of the cavity can be kept only in enamel, but for better retention and function, extension into the dentin is advisable. The total depth can be kept as 1.5 mm. The buccolingual width of the cavity is kept as small as possible (usual width is ¼th of the intercuspal distance).

The cavity walls may not be kept perpendicular and parallel.

The cavosurface margins are kept at right angle to the cavity walls, without my bevel. The butt joint preparation minimizes marginal breakdown and ditching of the composites.

The pits and fissures on the occlusal two-thirds of buccal and lingual surfaces of posterior teeth and lingual pits of anterior teeth can be restored with composites in a similar way. Butt joint preparation is preferred; however, a short bevel can be given for merging the color of composites. The depth of such preparation is 1.0–1.5 mm (as shallow as possible).

Class II Cavity

The occlusal part of the cavity is prepared as described for Class I cavity. The proximal box is prepared keeping in mind the extent of caries, convexity of the proximal surface, caries index and the potential masticatory load.

The procedure followed for composites is as for conventional box formation. The buccal and lingual walls of the proximal box are kept slightly away from the contact area (might not be in self-cleansing areas). The occlusal convergence, though not indicated, can be given to achieve better retention.

Modifications in Class II preparations

i. Minimal occlusal part is involved along with the proximal box (conventional) (Fig. 19.9A, B).
ii. Only proximal box is prepared without grooves; the occlusal part is not involved. Grooves are not given (Fig. 19.10A, B).
iii. Only proximal box preparation with retentive grooves (Fig. 19.11A, B).
iv. Preparation with unsupported proximal enamel (Fig. 19.12A, B).

A short bevel can be placed on the cavosurface margins of the walls of the proximal box for merging of color of composite restoration.

Class III Cavity

The cavity preparation should be initiated from the lingual side; in rare instances, labial approach is preferred (Fig. 19.13A, B)



Fig. 19.8: Class I cavity preparation (conventional) for composite in mandibular molar

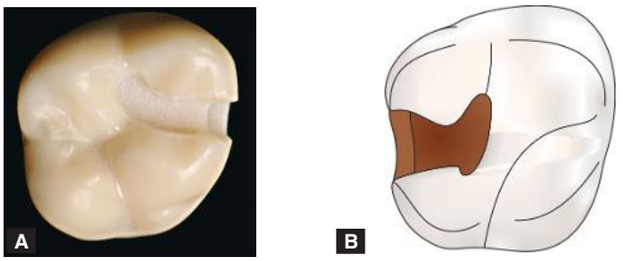


Fig. 19.9: Conventional conservative cavity preparation for composites: **(A)** On model; **(B)** Diagrammatic

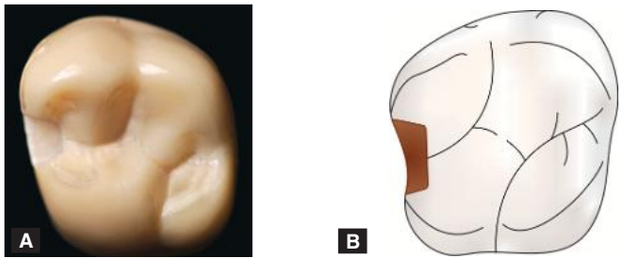


Fig. 19.10: Box only preparation without retention grooves: **(A)** On model; **(B)** Diagrammatic

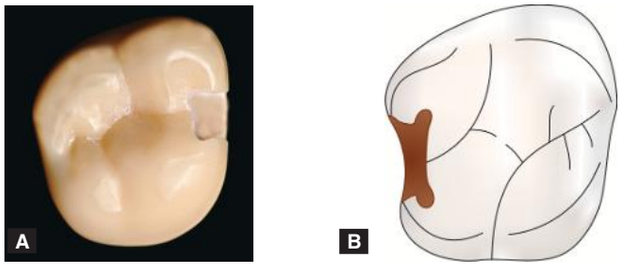


Fig. 19.11: Box only preparation with retention grooves: **(A)** On model; **(B)** Diagrammatic

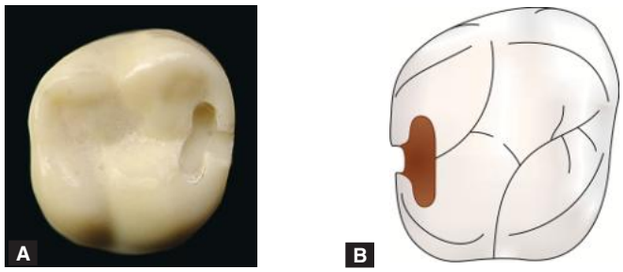


Fig. 19.12: Preparation with unsupported proximal enamel: **(A)** On model; **(B)** Diagrammatic

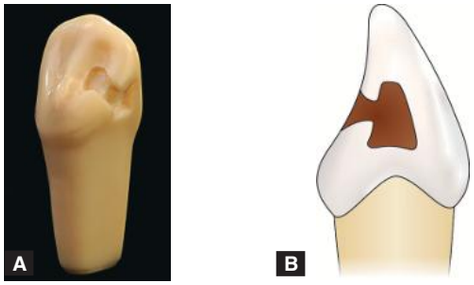


Fig. 19.13: Class III cavity preparation: **(A)** On model; **(B)** Diagrammatic

The cavity is prepared, in routine, from the lingual side, using conventional technique.

The depth of the axial wall should be 0.5 to 1.0 mm.

Usually, the depth is more incisally and as we move gingivally, the depth is decreased. The axial wall depth can be increased, in case where the gingival wall extends into the cementum, i.e. on the root surface. Sufficient axial depth is required for proper retention.

In case of labial approach, the outline form varies slightly. The cavity preparation is similar to the one described earlier with lingual approach; only difference is that a bevel is placed on labial cavosurface margin. This bevel is continuous from incisal to gingival side. The bevel also facilitates merging of color of restoration with the enamel.

Indications for labial approach are:

- Labial enamel is involved.
- Malaligned teeth, where lingual approach is difficult.
- Rotated teeth, where lingual side is hidden.

Class V Cavity

The outline form is dictated by the extent of the lesion at the cervical areas. The lesion can be limited to enamel only or may extend gingivally into cementum. The cavity is prepared by conventional means following the basic principles (Fig. 19.14A, B).

The occlusogingival width of the cavity is kept as minimum as possible, usually not exceeding 1.5 mm. In case the extension of the lesion dictates, the width can be increased accordingly (Fig. 19.15).

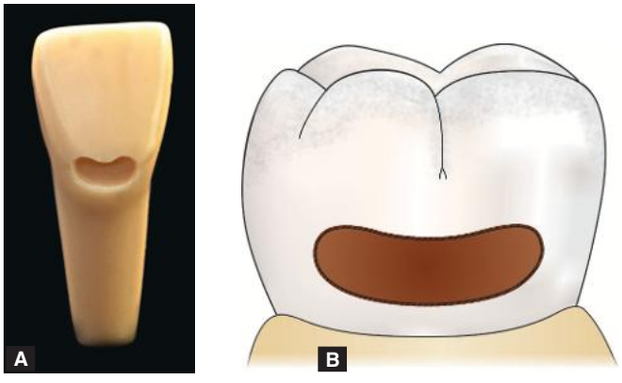


Fig. 19.14: Class V cavity preparation: **(A)** On model (anterior tooth); **(B)** Diagrammatic (posterior tooth)

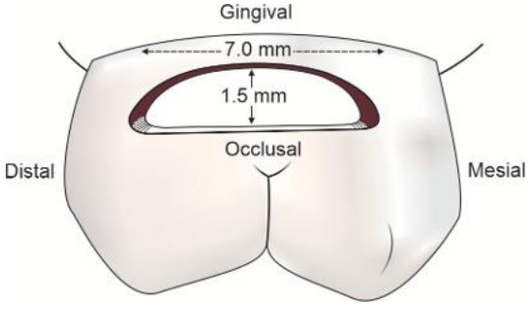


Fig. 19.15: Configuration of Class V cavity

The walls are finished with the help of chisel or marginal trimmer. The cavosurface margin of the walls is kept at 90°; except the occlusal wall, where a bevel is placed to have better merging of composite color with the tooth.

Retention grooves are not required in composite restorations. In extra large cavities, bilayered (sandwich) restorations are preferred.

Modification of Class V cavities: The Class V cavity preparation is modified depending upon the location and extent of the lesion.

a. When the lesion is on the line angles of buccal/lingual walls of posterior teeth (Fig. 19.16), such lesions, as mentioned earlier, have not been included in Black's classification. The outline form depends upon the size of the lesion. The usual form given is rounded or oblong and the cavity is prepared as described for class V cavity. The short bevel can be placed, wherever the operator feels the necessity of color merging with the tooth.
b. In rare instances, cervical lesions extend occlusally along the sides. This extension can be unilateral or bilateral (Figs 19.17A, B and 19.18A, B). The side of the occlusal wall is raised accordingly and merged with the main cavity preparation. The bevel can be placed all along this wall.

Class VI Cavity

The outline form is dictated by the extent of caries. Usually, these lesions are small and not very deep.

The depth of the cavity is maintained inside the dentinoenamel junction. The surrounding walls should have definite dentin support (Fig. 19.19A, B). The shape of the cavity is usually round in posterior teeth and squarish in anterior teeth. The cavosurface angle is usually 90° and bevels are not preferred.



Fig. 19.16: Cavity on the line angle

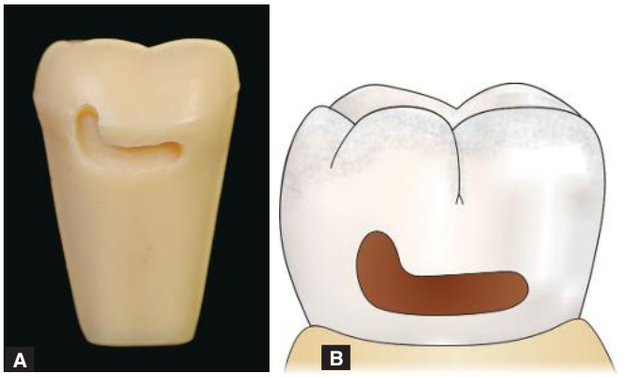


Fig. 19.17: Class V cavity with unilateral extension: **(A)** On model; **(B)** Diagrammatic

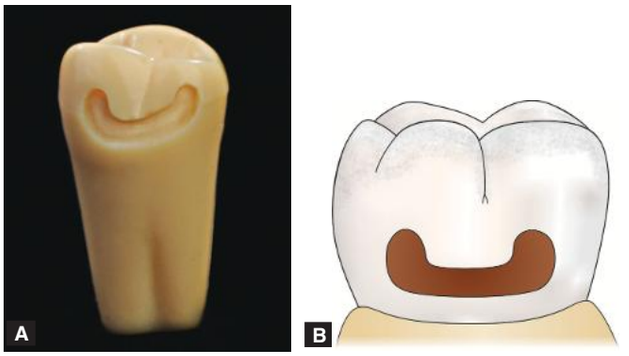


Fig. 19.18: Class V cavity with bilateral extension: **(A)** On model; **(B)** Diagrammatic

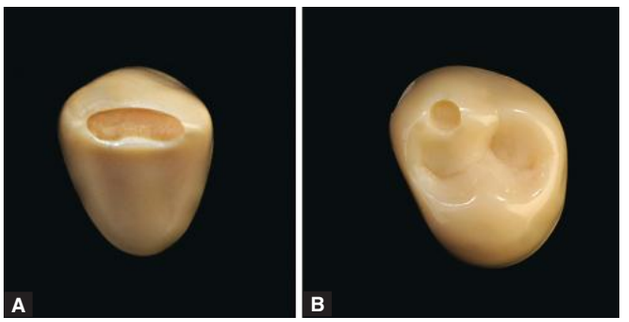


Fig. 19.19: Class VI cavity on **(A)** mandibular canine; **(B)** Maxillary premolar

PLACEMENT OF COMPOSITES

The placement of composite in the prepared cavity is carried out following the sequence as follows.

a. Pulp Protection

After the cavity preparation is completed and the remaining caries is excavated, the depth of the cavity inside the dentinoenamel junction is thoroughly evaluated. In deep preparations (remaining dentin is less), the pulp is protected by applying base/liner.

Calcium hydroxide and glass-ionomers are preferred bases under the composite restorations. Zinc oxide eugenol is contraindicated, since it hinders with polymerization of methacrylate group of composite resins. Light cure calcium hydroxide, and glass-ionomer cement are preferred, since over these surfaces, etching is not required. Compomers are also being used as base under the composites. Glass-cermet is a preferred base, especially at the cervical wall of the proximal lesions. The release of fluoride ions at the cervical wall minimizes the chances of secondary caries at these areas; and the adequate strength of glass cermet cement effectively hold the restoration.

b. Acid Etching

Isolate the tooth properly and make the area accessible for acid etching. Gel etchants are preferred over liquid etchants, since overflowing is common with liquid etchants. Liquid etchants can be used, where the flow is not towards the gingival side (labial side of upper anterior teeth). The liquid etchants should not be used on the mandibular teeth, since the liquid may trickle into the gingival crevices.

The gel etchants can be applied with the help of plastic instrument or sprayed with the fine needle provided with the syringe. The etching time depends upon the area to be etched, following the manufacturer's instructions (etching time is doubled in deciduous enamel).

After the stipulated time (usually 10–15 seconds), the etchant is washed thoroughly. Care should be taken not to leave any residual acid on the cavity walls. The area is dried preferably using chip syringe (*compressed air should be avoided as it may injure the enamel tags*).

The etched enamel will appear clean, white and frosted. Do not touch the etched surface with any instrument, cotton or hand (Figs 19.20 to 19.23). In case saliva or gingival fluid touches the etched surface, repeat the procedure again.

c. Applying Bonding Agent

The bonding agent is applied over the etched surface with the help of brush or the tip supplied by the manufacturers. The bonding agent is made to flow uniformly all along the walls with the help of air from chip syringe. Air should be gently blown without any pressure, otherwise there might be no bonding agent at certain areas of the cavity. Then it is light cured for 10 seconds.

Complete isolation should be maintained in case saliva and/or gingival fluid trickle in, the whole process is to be repeated right from etching.

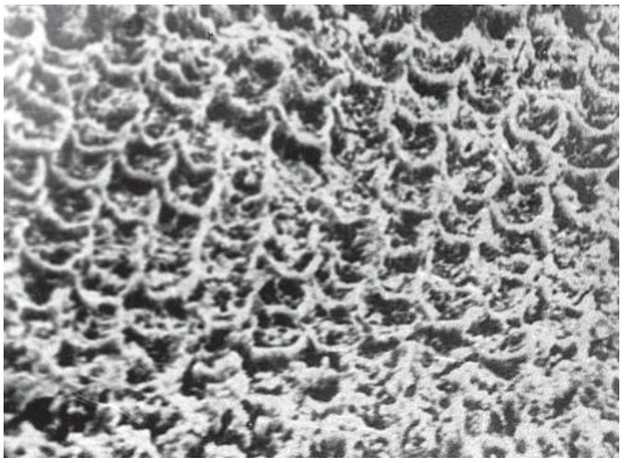


Fig. 19.20: Enamel surface after etching

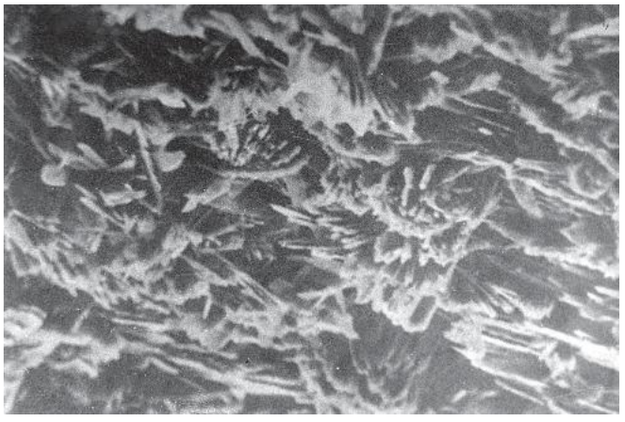


Fig. 19.21: Etched enamel without washing

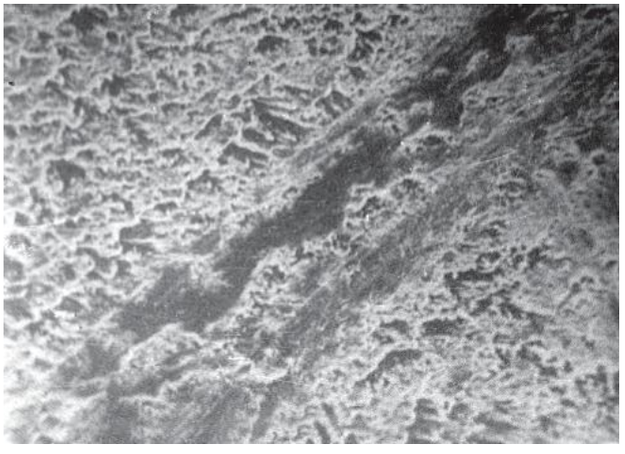


Fig. 19.22: Etched enamel after touching with cotton

A few authors advocate applying bonding agent to the cavity walls and gently rubbing it, facilitating its penetration into dentinal tubule area. A second coat of bonding agent is applied over the first layer to achieve universal bonding in the entire area.

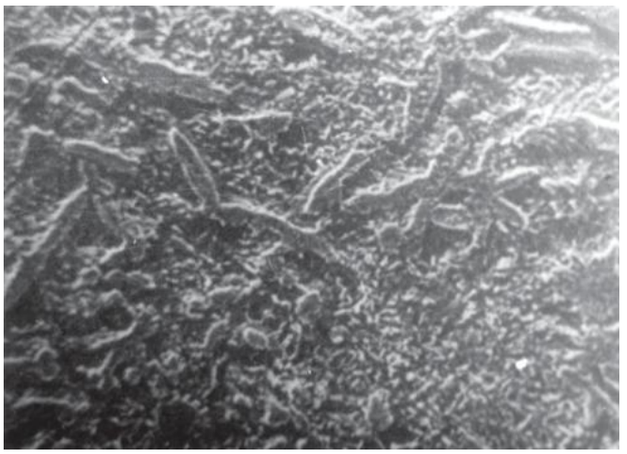


Fig. 19.23: Etched enamel when contaminated with saliva

d. Placement of Matrix

Matrices are placed to achieve the proper contact and contour of the tooth at the proximal area.

The placement of matrix after the etching and bonding steps is preferred. In case the matrix is placed before these steps, the application and removal of etchant become difficult.

After placing the matrix, it is stabilized with the help of wedge or impression compound, as the need be. Details regarding matrices and wedges are given in Chapter 11.

Two types of matrix are usually used for composite restorations:

i. Metal matrix

ii. Polyester (transparent) matrix

The metal matrix (0.001–0.002 inch) is shaped outside the oral cavity with the help of handle of the mirror or with contouring pliers. It can be suitably cut to suit the size and contour of the tooth. The matrix should be extended beyond the marginal ridge and the gingival area. Care should be taken not to impinge the gingival tissues. A properly shaped wedge is then inserted to make the matrix stable.

Wedge helps in the following ways:

- Stabilize the matrix
- Facilitate slight separation of teeth
- Prevents overhangs at gingival areas
- Creates gingival embrasures effectively

Polyester (transparent) matrices, which can reflect light, are also used for proximal restorations. With the introduction of fine curing tips and the light reflecting wedges, proper curing of composite at gingival third area has become easier. In case the transparent matrix is applied, it is exchanged with metal matrix after completing the gingival curing.

e. Restoration

Syringes with composites (usually flowable composites) are available with disposable needles to be applied directly in the cavity. Guns with ampoules of composites are also available in various sizes and shapes (Fig. 19.24).

Basically, guns are used for viscous composites and the syringes for flowable composites.

Usually, Teflon-coated instruments are used to pick and insert the composite (Fig. 19.25). Stainless steel instrument should not be used, since the material will stick to the instrument. The instrument should be kept dry during manipulation and be wiped with cotton at every pick. In no case, the instrument should be touched with hands or any moist object.

i. *Restoring Anterior Teeth (Class III & Class IV Cavities)*

The composite in small increment is placed first at the gingival wall. (Whether the approach of cavity preparation is labial or lingual, the gingival wall is first to be restored.) Use of retraction cords and pushing of rubber dam with plastic instruments can help visualize the gingival cavosurface area. The first increment should be kept small, approximately 1.0 mm thick and the subsequent increments of



Fig. 19.24: Composite gun with ampoules of composite

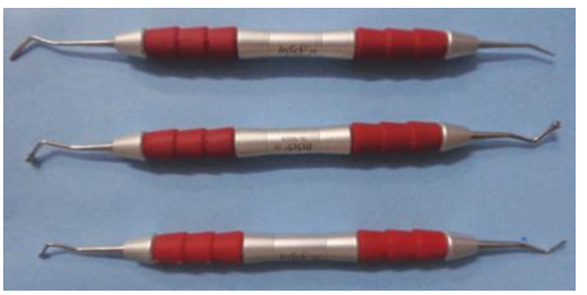


Fig. 19.25: Teflon-coated instruments for manipulating composite

1.0–1.5 mm thickness can be applied over the first. Each increment is cured for 20 seconds and final one cured for 40 seconds to one minute. If lingual approach is utilized, curing is preferred from labial aspect and vice versa. Finally, the composite should be cured from all the surfaces. The stepwise restoration of fractured incisal portion is shown in Fig. 19.26A to I.

Blending of shades of composite is important. From cervical to incisal area, the color of composite would be from dark to light to translucent (darker shade at the cervical, less dark in the middle, lighter at the incisal and translucent at the incisal edge).

ii. *Restoring Posterior Teeth (Class II Cavities)*

The restoration is initiated keeping in mind the volume of the cavity and the areas, which would come under stresses. The volume of the cavity will dictate the operator regarding number and locations of the increments and the future stress-bearing areas will dictate the operator to use particular type of composites. The stepwise restoration of proximal lesion is shown in Fig. 19.27A to F.

The cavity for composite is always restored in increments. The increments can be placed in a variety of designs, but the thickness of the increment should be kept as small as possible, i.e. not exceeding 1.5 mm at a time. Each increment is cured for 20 seconds before placing the next over it. A few authors are of the view that partial curing of these increments would lead to better adaptation of each increment. This phenomenon is known as '*soft start polymerization*'. (Initial increment is cured for 10 seconds before placing the next.)

The different designs of increment placement are:

i. *Three increment design:* One flat increment at gingival and pulpal and two oblique increments both at proximal box and the occlusal box (Fig. 19.28A).
   The first increment (may be 1.0 mm) is always cured from the sides, rather from the occlusal end. This is the simple and accepted design.
ii. *Horizontal layering design:* Small increments are placed horizontally one above the other, starting from gingival wall to occlusal end. The number of layers depends upon the depth of the proximal box (Fig. 19.28B).

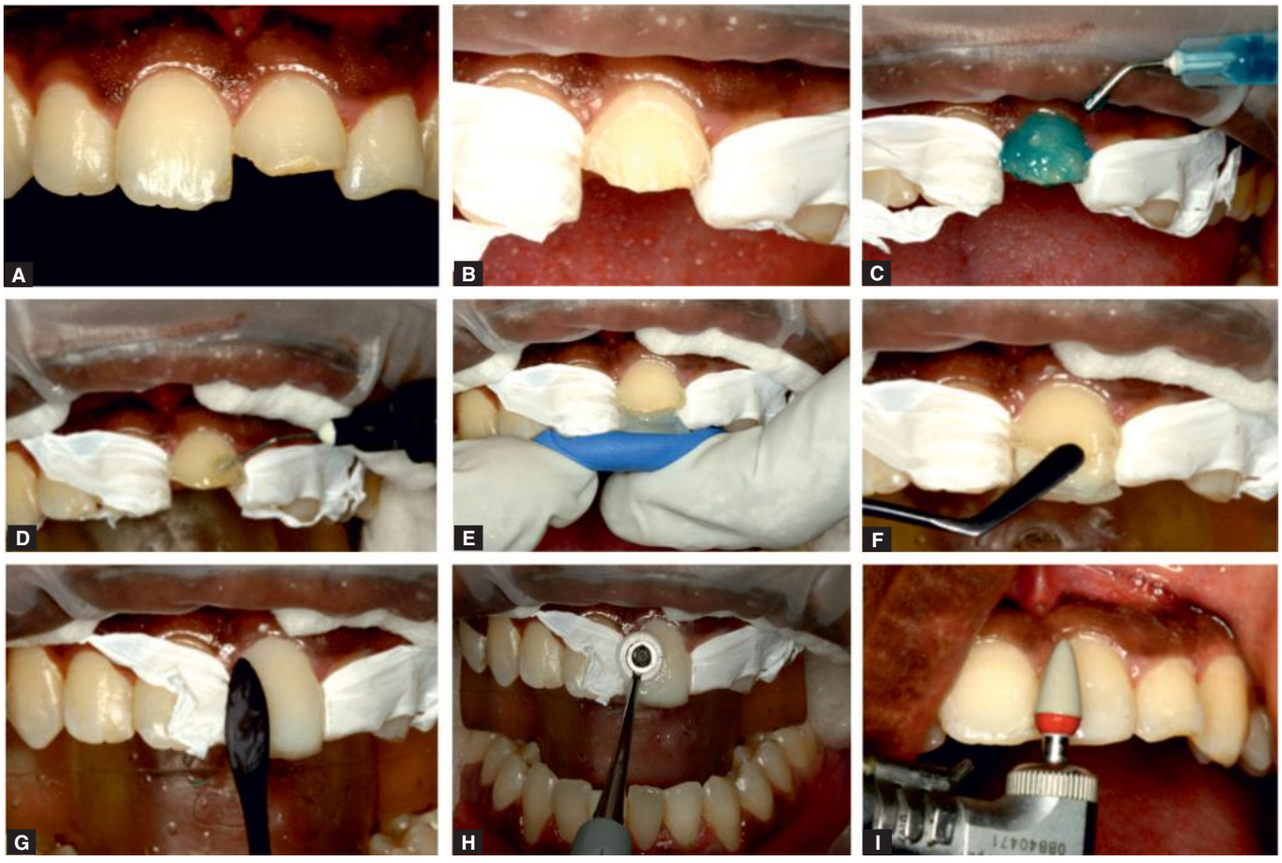


Fig. 19.26: Restoration of fractured incisal portion: **(A)** Preoperative; **(B)** Isolating the tooth; **(C)** Etching the surface; **(D)** Preparing mold; **(E)** Fixing mold with the help of impression; **(F)** Building composite; **(G)** Shaping the restoration; **(H)** Finishing with restoration; **(I)** Final polishing

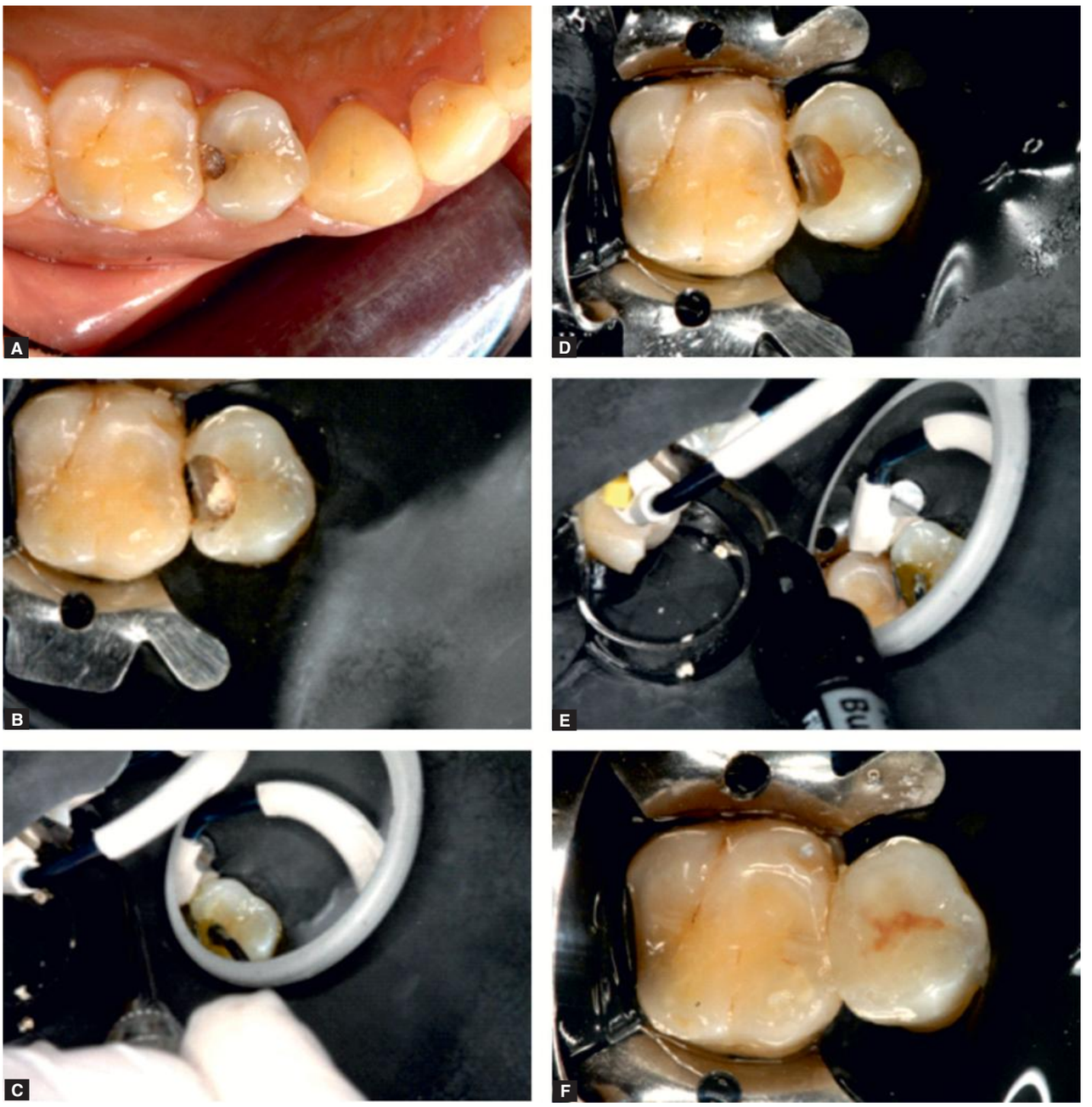


Fig. 19.27: Restoration of proximal lesion: **(A)** Preoperative; **(B)** Cavity preparation; **(C)** Placement of base; **(D)** Processing composite placement; **(E)** Restoring the cavity; **(F)** Postoperative (final restoration)

iii. *Oblique layering design:* Each increment is placed obliquely starting from any side, i.e. buccal or lingual (Fig. 19.28C).
iv. *U-shaped layering design:* At the base (both gingival and occlusal) U-shaped increment is placed first, followed by horizontal and oblique increments over the first increment (Fig. 19.28D).
v. *Vertical layering design:* The increments are placed in vertical fashion starting from one wall, i.e. buccal or lingual and carried onto other wall in small increments (Fig. 19.28E). The curing is started from behind the wall as the ease may be, i.e. if first increment is placed on the buccal wall, it is cured from outside the buccal wall.

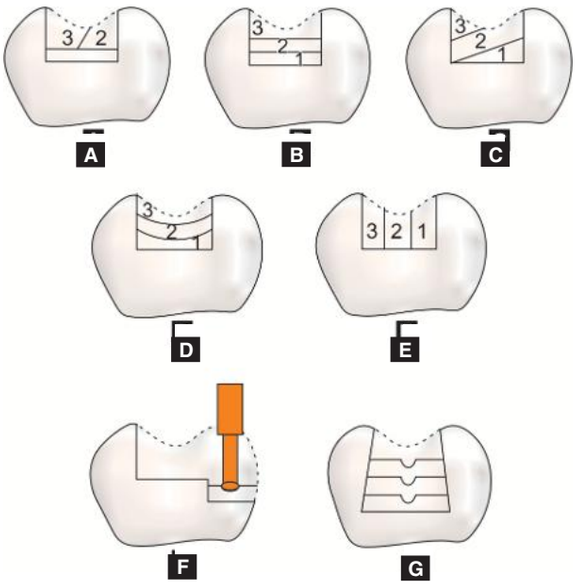


Fig. 19.28: Different designs of increment placement for composite: **(A)** Three increment design; **(B)** Horizontal layering design; **(C)** Oblique layering design; **(D)** U-shaped layering design; **(E)** Vertical layering design; **(F & G)** Curing each increment by inserting the fiberoptic microtip into the composite

vi. Layering technique in the proximal box and curing each increment by inserting the fiberoptic microtip into the composite (Fig. 19.28F, G).

The increments are compressed with Teflon-coated instruments, so as not to have any voids between the two increments. Use of bonding agent in between the increment is not required, since partially cured increments unite of their own. However, a few authors advocate use of bonding agents after the last increment to have better marginal adaptability.

Establishing Proximal Contact

One of the problems with composite restoration is establishing proper contact at the proximal sides. Since composites shrink after setting, coupled with the gap created by matrix, this problem increases two-folds. Though with the advent of newer composites and newer techniques along with ultrathin matrices, the problem is minimized, yet it is not completely solved.

Proximal contacts are restored in two ways:

i. After curing of the gingival increment up to the contact area, the wedge and the matrix is removed and rest of the restoration is carried out without the matrix. A few authors are of the view that the minor separation created by the matrix placement and wedging is sufficient for the establishment of final contact, therefore, no need to remove matrix.

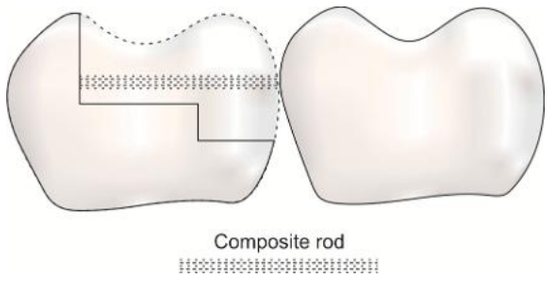


Fig. 19.29: Composite rod and its placement at the desired place adjacent to the contact area (Duokoudakis method)

ii. Duokoudakis (1996) devised a technique whereby proper contact can be established at the proximal areas. According to him, the width of the cavity preparation from the contact area to the opposing cavity wall is measured. The area of the contact, which is to be restored, is also measured (Fig. 19.29).

A small tube, containing composite of the same width and slightly more in length is taken and cured. This composite rod will be slightly larger in size and finally adjusted by finishing at the margins (Fig. 19.29). The composite is restored as in routine keeping the matrix and the wedge in place and the rod is kept at the desired place adjacent to the contact area (Fig. 19.29). Finally, the rest of the cavity is restored with the composite. This technique helps in maintaining the proper size and shape of the contact.

FAILURES IN COMPOSITE RESTORATIONS

Composite restorations are very technique sensitive; care should be exercised at each step of composite placement to minimize failures. The visible modes of failures in composite restoration are:

- Discoloration, and marginal ditching (Fig. 19.30).
- Fracture of restoration (Fig. 19.31).
- Recurrent caries (Fig. 19.32).
- Lack of maintaining contact.
- Accumulation of plaque around the restorations
- Postoperative sensitivity

Features leading to failure in composite restoration are:

a. ***Incomplete excavation of caries and zinc oxide:*** The caries if not thoroughly removed (Fig. 19.33), the left over caries may hinder with the bonding mechanism. Similarly, if zinc oxide eugenol is not removed thoroughly from the base of the cavity and the walls, this will also lead to failure (Fig. 19.34).

b. ***Failure to maintain etched surface:*** Repeated touch after etching, blowing with compressed air under pressure and/or sweeping the etched surface with cotton can lead to fracture of enamel tags, leading to lack of proper union between enamel tags and



Fig. 19.30: Marginal staining and ditching around composite restoration



Fig. 19.33: Caries not thoroughly removed during cavity preparation



Fig. 19.31: Fractured composite restoration



Fig. 19.34: Failure to remove residual zinc oxide filling before restoration



Fig. 19.32: Recurrent caries adjacent to composite restoration

the resin. Many a times, the acid is not washed out thoroughly and the residual acid in the tags hinders with the bonding. For drying, air from chip syringe should be used. Three-way syringe should be avoided as the air could be contaminated by machine oil or water.

c. ***Non-uniform application of bonding agent:*** Bonding agent is to be applied uniformly all around the cavity walls. Non-uniform bonding hinders the union of material with the bonding agent. Excess of bonding may facilitate marginal leakage at a later stage (Fig. 19.35).

d. ***Lack of isolation:*** A rubber dam isolation is mandatory; however, cotton rolls can be used, if the time consumption is less (Fig. 19.36). Gingival retraction cords can be utilized in Class III and Class V cavities and also in composite laminates. The touch of saliva and the gingival fluid is to be avoided even after etching and bonding phenomenon. In patients with uncontrolled salivation, antisialagogue drugs can be used.



Fig. 19.35: Excess bonding agent applied in the cavity preparation



Fig. 19.37: Composite being touched with fingers



Fig. 19.36: Lack of isolation during restoration



Fig. 19.38: Composite fallen on the tongue

e. ***Touch of composite with fingers:*** The composite should not be touched with fingers (Fig. 19.37). Many a times, the operator uses his finger to pack or keep the material in place. Always pick and hold the composite with Teflon-coated instruments. Sometimes the composite falls over the tongue (Fig. 19.38) and the operator picks it up and pack the same into the cavity. This will lead to failure.

f. ***Avoid bulk placement of composites:*** The composite is to be placed in increments (Fig. 19.39) and each increment should be as small as possible, since the contraction after polymerization leads to gaps at the tooth–restoration interface. These gaps consequently lead to postoperative sensitivity, marginal leakage and secondary caries.

g. ***Improper curing:*** The curing of the composite should be from all the sides and for stipulated period. The filter and the bulb vis-à-vis the intensity of light should be checked in routine to ensure proper curing. For proximal restoration, fine curing tips can be used.

h. ***Improper finishing and polishing:*** Composites should be finished and polished thoroughly.



Fig. 19.39: Composite being placed in bulk

Improper finishing/polishing will invite plaque accumulation and lead to surface staining. The occlusion should be checked before final polishing. Antagonist tooth, if impinging, can be grinded off.

Ceomposite restorations: Clinical guidelines

- Cavity preparation should be kept as small as possible.
- Avoid sharp internal line angles in cavity preparation, which increases stresses.
- Deeper cavities should be given base of calcium hydroxide or glass-ionomer cement.
- Follow strict isolation regime.
- Avoid inadequate curing; may lead to hydrolytic breakdown of composites.
- Preferably restore in small increments.
- Fill proximal box separately and create proper contact areas.

COMPOSITE LAMINATES AND VENEERS

The laminates and veneers are interchangeable terms; the only difference is in laminates, the overall shape is not changed, while in veneers the shape of the tooth is changed. Laminates are given only where the teeth are discolored but their shape, size and position in the arch is normal; whereas, veneers are given where the teeth need change in their shape, size and the position in the arch.

The laminates are preferred in maxillary anterior teeth since these teeth play a predominant role in smile and esthetics of a person. Mandibular anterior teeth are rarely laminated because of lack of space and more so, the teeth are hidden under the lower lips.

Composite laminates can be direct and indirect depending upon whether the laminate is fabricated inside or outside the oral cavity.

Indirect laminates are advantageous since they provide better esthetics, less fatigue to the patient and easily manipulated. However, the procedure is completed in two steps. Direct laminates, though save patient's visits, are tiring and also technique sensitive.

a. Direct Laminates/Veneers

Tooth Preparation

Preparation varies depending upon whether the discoloration envelopes the marginal ridges and the incisal edge or not. In deep dark stains, which extend towards the incisal enamel, the preparation involves the incisal edge. Otherwise, the preparation ends slightly away from incisal edge. Similarly whether to extend the preparation below the gingival crest is also dictated by the extent of the stain at these areas. Preferably, the preparation should be kept 0.5 mm supragingivally to avoid any impingment to the gingival crests.

The tooth is prepared using fissure burs. The depth of the preparation is 0.5 to 0.7 mm and is kept in enamel only. In case, the underlying portion is dark, further cutting is initiated, total depth being 1.0 to 1.25 mm. The extra depth is utilized for the opaquer to mask the dark color.

A few authors advocate use of cross-cut burs for cutting labial enamel, so as to achieve undercuts and large surface area for etching. Though with this technique, retention is increased, but utmost care is necessary not to overcut the tooth.

Placement of Composite

After shade selection, the tooth is isolated using rubber dam. Rubber dam is preferred, since laminates are time consuming and the cotton rolls will not be effective for such a long time.

In case, the preparation extends beyond the gingival crest into the subgingival area, use of retraction cords is mandatory.

The area is etched for 15–20 seconds, if the preparation is in enamel, and 10 seconds, if it is in dentin. The etchant is thoroughly washed and dried using air from chip syringe. A thin coat of bonding agent is applied, made uniform by blowing air without pressure and cured for 10 seconds. The non-uniform application of bonding agent should be avoided.

The selected composite is placed in small increments from cervical area to the incisal. Blending of color from one shade to other is very important. Two shades can be mixed and applied at the junction of cervical-middle third areas; and also at middle-incisal third areas. Slightly dark shades are required at the line angles. Finally, the incisal edge is restored using translucent composite. Coloring tints (available in various shades) can be used to have the reflection of different shades. In case the tooth is darkly stained and the preparation is in dentin, the middle area of the preparation or the 'body' of the preparation should be filled with opaquer.

A slight overcontoured tooth is preferred at this stage, since at the time of final finishing, the exact contour can be re-established. In case, only one tooth is laminated, the final contour can be checked with adjacent tooth and if more than one tooth is to be laminated, overall patient's smile requirements and esthetics should be kept in mind (Fig. 19.40A, B).

Illusion

The term 'illusion' is coined with two words 'illumination' and 'vision', i.e. what is visible to eyes with the reflection of light. The phenomenon of illusion is utilized where the smile of the individual is to be modified. It is established that round incisal line angles, greater incisal embrasures and less prominent marginal ridges produce feminine smile; whereas,



Fig. 19.40: Direct composite laminates on maxillary central incisors: **(A)** Preoperative; **(B)** Postoperative

reverse is true for masculine smiles. For young ladies, narrower appearance of anterior teeth would be more pleasing while for adult males, wider anteriors are preferred.

All these features can be achieved; for example, in diastema closure the teeth appear to be widened; however, with the phenomenon of illusion, these can be made to appear narrower and more pleasing.

The reflection of light varies at different height and depths of the same object. In case we need narrowing of teeth, the mesial and distal marginal ridges are brought closer to each other and the light reflected over these areas will make the tooth appear narrower. Alternatively, to make the tooth appear wider, the mesial and distal marginal ridges are separated mesially and distally.

In case lengthening or shortening is needed, the gingival bulge area is kept more incisally and the incisal bulge is placed gingivally. This will produce an illusion of long teeth. Reverse is true for making illusion of shorter teeth by keeping the gingival bulge more gingivally and the incisal bulge more incisally.

b. Indirect Laminates/Veneers

The steps involved in preparation of indirect laminates/veneers are:

i. ***Preparation:*** The preparation of the tooth is almost similar to the one for direct veneers; the only difference is the incisal enveloping is not preferred in indirect veneers. The one path of withdrawal and insertion (labial always) should be followed with no undercuts in the preparation. The usual depth of the preparation is 0.75 mm and almost always in enamel. In case, the preparation is made deeper into the dentin, because of staining or any other reason, the area can be restored with opaque material before making the impression.

ii. ***Impression making:*** The gingival embrasures are filled with soft wax from the lingual side and the rubber base impression is taken. The impression is thoroughly checked under a magnifying lens for any air bubble, etc. and is poured in die stones. The positive replica is checked for any undercuts, etc. If any discrepancy exists, the procedure is repeated.

iii. ***Fabrication:*** The laminate is fabricated on the cast/die (positive replica). The composite material is placed in increments, taking care of the blending of shades from gingival area to the incisal and also in the mesial and distal line angles.

The composite is cured during placement of each increments and finally for 40 seconds. The curing is also carried out from inside for 40 seconds. A few authors advocate boiling the veneers in water for 20 minutes to achieve better curing and subsequently, the strength. Finally, the veneer is finished and polished and kept ready for fixation.

iv. ***Fixing:*** The prepared tooth surface is washed, cleaned and dried. The area is isolated. The enamel surface is acid etched for 20 seconds, washed and dried with oil-free air. Bonding agent in thin film is applied over the etched surface uniformally by blowing air gently from the chip syringe. The inner side of the composite veneer is also etched with weak acid. After placing the veneer, the margins are checked thoroughly evaluated; modified, if need be.

v. ***Finishing and polishing:*** Though the veneers are polished in the laboratory, minor finishing and polishing may be required at the margins. The occlusion, particularly the protrusive movement is checked to restore occlusal harmony. Patients are instructed not to try reverse bite with these veneers (Fig. 19.41A to C).

For details, refer Author's book *Indirect Restoration*, 2nd edition.

COMPONEERS

Componeers are pre-fabricated composite veneers (Fig. 19.42A), especially manufactured in thin shells



Fig. 19.41: Indirect composite laminates: **(A)** Preoperative (maxillary anteriors discolored); **(B)** Preparation for composite laminates; **(C)** Postoperative (restored laminates)

that can be fixed onto the labial surfaces of teeth. The procedure is easy, immediate and economical. The indications are the same as for other veneers, such as correcting minor defects, gaps, tooth discoloration, broken edges, etc. The advantages are improved esthetics in one visit, customized shaping of front teeth and economical. The shine of the veneer can be refreshed by polishing. The componeers are cemented using the same cementing agent, thereby improving bonding.

Procedure

The procedure involves the following steps:

- The rubber dam sheet is cut from second premolar (15) to second premolar (25) and the clamps are fitted on the 1st molars (16 and 26) on both sides (use of split rubber dam) (Fig. 19.42B). The loose ends on the dam are attached to the mucous membrane area with the help of butyl cyanoacrylate.
- *Enamel preparation:* Minimal tooth preparation is required for componeers. (Thickness of componeer is 0.3 mm at the cervical margin, which increases to 0.5 mm at the middle and 0.7 mm at the incisal edge.) Only roughening of the enamel is enough for patients who agree for a little protrusion, but if it is a single tooth component, where arch form has to be maintained, the 0.3 to 0.5 mm of reduction is required (Fig. 19.42C). Interdental areas are also roughened to accommodate the thickness of composite. A diamond-coated single-sided re-contouring strip is required for interproximal preparation of enamel.
- *Size selection:* The size of the componeer is selected with the help of contour guides provided in the kit (Fig. 19.42D). The contour guides are transparent and light blue in color to complement the color of the tooth. The rule is that the componeer should cover the width of the tooth and if the componeer is longer cervicoincisally, it should be adequately trimmed (Fig. 19.42E).
- 37% phosphoric acid etching (Fig. 19.42F) is carried out for 15 seconds, followed by rinsing for 10 seconds.
- After drying the tooth surface, appropriate gingival retraction cord is placed in the free gingival margin of the prepared teeth.
- Two layers of appropriate bonding agent is applied to the prepared teeth (first one is rubbed for 15 seconds with nylon tufted brushes). After the application of 1st layer and air blowing it, mylar strips are placed between the teeth and then the second layer is applied (Fig. 19.42G). The bonding agent is light cured for 20 seconds lingually and then 20 seconds buccally. The same bonding agent is applied on the inner surface of the componeer, which is held with the help of componeer holder (provided in the kit) and air blown. (This layer is not light cured.)
- The chosen shade of composite is applied on the inner surface of componeer and the same composite is applied on the bonded tooth surface.
- The componeer is compressed on the tooth surface and pushed into its final position with the help of componeer placer, provided in the kit (Fig. 19.42H). The extra composite (flash) is removed with the Instrument provided in the kit (Fig. 19.42I). Finally it is light cured for 30 seconds both lingually and buccally.
- The composite in the interdental areas is removed with fine tapered bur (Fig. 19.42J). The inter-dental finishing is carried out with finishing strips (Fig. 19.42K). The restoration is polished with appropriate polishing agents (Fig. 19.42L). The restored tooth is checked for occlusal disharmonies (Fig. 19.42M).

Advantages

- Minimal invasive and conservative tooth preparation
- No impression required
- Minimum application time
- Cementation using the same material
- Highly polished surface (esthetically pleasing)
- Cost-effective

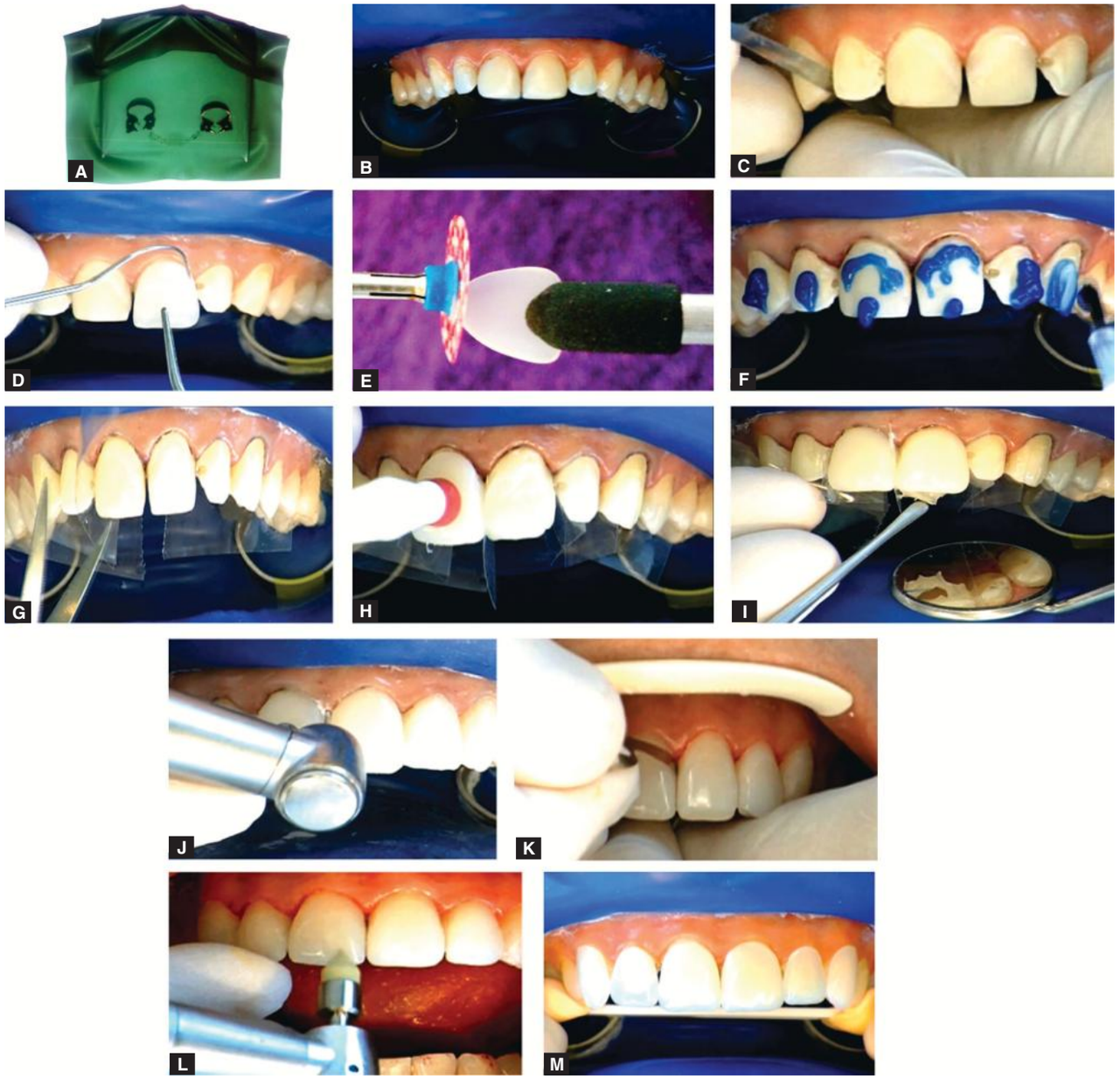


Fig. 19.42: Placement of componeers: **(A)** Componeer kit; **(B)** Rubberdam application; **(C)** Enamel preparation; **(D)** Selection of componeer; **(E)** Minor adjustment; **(F)** Etching; **(G)** Adjusting Mylar strip; **(H)** Compressing the componeer; **(I)** Removing excess flash; **(J)** Gross finishing of interdental areas with burs; **(K)** Finishing interdental area with finishing strips; **(L)** Polishing; **(M)** Occlusal checking

Limitations

Two bonding interfaces are involved; one, the inner surface of the processed composite resin veneer and second, the acid-etched enamel. Tensile strength ranges from 2000–2750 psi.

The degree of polymerization is significantly greater with indirect resin veneers than that achieved by chairside conventional polymerization techniques; therefore, the potential for establishing strong chemical bonds between the processed veneer and resin-bonding medium is significantly diminished.

The preparations may involve considerable removal of sound tooth substance. The inherent nature of composite resins exhibiting polymerization shrinkage, dimensional changes, staining and poor wear resistance may also lead to poor prognosis.

The microfilled composite resins can achieve satisfactory appearance and shine. However, microfilled composites wear at twice the rate of

conventional composites as a result of tooth brushing. To overcome this problem, it is suggested that conventional or hybrid composite should be used to form the bulk of the veneer and a thin layer of microfill composite be placed on the surface for appearance and surface shine. This superficial layer would have to be replaced at intervals. The acrylic veneers could overcome some of these problems; however, the restored teeth being bulky, coupled with weak chemical bond between cement and veneer, may lead to failure.

Both composite and acrylic veneers have a dull, monochromatic appearance. Low abrasion resistance leading to loss of contour and staining, and loss of surface luster, biological incompatibility with gingival tissue, limited bond between laminates and composite, were the major drawbacks of these resin veneers.

COMPOSITE INLAYS

A composite inlay is defined as "*a restoration fabricated with composite material and is cemented into a cavity prepared specifically for the composite*".

Need for a Composite Inlay?

Direct posterior composite restorations have certain limitations, viz.

- Despite the incremental build-up techniques, polymerization shrinkage is a major problem, which contributes to marginal defects, cuspal distortion, crack formation and propagation within the tooth tissues, resulting in postoperative sensitivity.
- Inadequate polymerization in deep interproximal areas; may lead to partly cured restoration.
- Difficult to achieve proper bond with dentinal surface.
- Increased wear in load-bearing situations.
- Water sorption with resultant hydrolytic instability.
- Contour, contacts are not proper.

To overcome these clinical problems, the indirect composite inlay system was introduced. It has been established that inlays enhance the hardness and wear resistance of the material and also reduces the microleakage providing better adaptability at the tooth–restoration interface.

Indications

- Conventional tooth-colored restoration of better quality.
- Moderate- to large-sized lesions, where sufficient tooth tissue remains following preparation.
- Where there is no evidence of excessive tooth wear.
- Restoration not overloaded occlusally.
- When better control over contacts and contours is desired.

Contraindications

- Inadequate remaining tooth tissues to create proper resistance and retention forms for the purpose of bonding.
- Teeth, which show excessive wear.
- Situations, where moisture control cannot be secured and maintained.
- Teeth that experience heavy occlusal forces, like in bruxism, clenching, etc.
- Un-cooperative patients.
- Patients who do not maintain oral hygiene.

Advantages

- Better compressive strength, tensile strength and hardness
- Better viscoelastic stability
- Increased creep resistance to occlusal stresses
- Better control of contacts and contours
- Improved marginal adaptation
- Soft and gentle to nearby tooth
- Less technique sensitive
- Saves chair time.
- Can easily be adjusted in the oral cavity
- Easy manipulation in the oral cavity

Disadvantages

- Fabrication requires two appointments and a temporary restoration.
- Preparation of the cavity may necessitate removal of sound tooth structure to create parallelism and other features.
- Temporary restoration, if not proper, may lead to bacterial contamination, etc.
- Durability is not established.
- In case fitting is not proper, luting agents do not help.

Classification of Composite Inlays

Composite inlays can be classified on the basis of curing protocols as:

a. *Heat cured:* The inlays are cured at elevated temperature and under pressure in one stage. The composite employed is heat cured rather than light cured, e.g. SR Isosit system. (Inlays are cured at 120°C and under pressure.)
b. *Light cured:* One mode of curing (Inlay is cured by light only.)
c. *Dual cured:* Initial curing at room temperature by light followed by additional curing by heat and light, e.g. Coltene Brilliant system.

Evolution of Composite Inlay Material

The evolution of composite inlay system is categorized as 'Generations'.

a. First Generation Composite Inlays

i. ***SR Isosit system:*** SR Isosit (microfilled composite) is a homogeneously filled composite containing 55% by weight colloidal silica plus 20% lanthanium fluoride. The composite is available in seven shades.

ii. ***Coltene Brilliant system:*** Coltene Brilliant system is fine hybrid composite containing 78.5% glass fiber (particle size 0.5 nm). The composite inlay is cured on a fabricated die and further polymerized in the Coltene DI500 light/heat curing oven for 7 minutes at a temperature of 120°C. The composite is available in four vita shades.

iii. ***Kulzer inlay system:*** Kulzer system is coarse hybrid composite containing 80% glass ceramics. The composite inlays are light cured and tempered for 3 to 6 minutes in light units.

iv. ***Visio Gem:*** Visio Gem inlays are initially light cured, followed by a 15-minute curing under vacuum.

b. Second Generation Composite Inlays

i. ***Art glass:*** Art glass contains barium glass and a moderate amount of colloidal silica. The material is photocured in a special unit using a xeno stroboscopic light (emission of 4.5 watts power in the range 320 and 500 nm). This range is significant because excitation of camphorquinone (initiator) is optimized at about 470 nm.

ii. ***Belleglass HP and NG:*** Belleglass HP is polymerized under pressure at an elevated temperature (138°C) and in the presence of nitrogen. The use of nitrogen during polymerization helps increasing the wear resistance of the material. Nitrogen produces an oxygen-free environment, which results in higher level of polymerization. Belleglass NG incorporates nanoparticles (submicron filler particles), providing improved physical properties.

iii. ***Tescera:*** Tescera utilizes microhybrid composite; in addition to heat and light, air pressure is used for polymerization.

c. Third Generation Composite Inlay

i. ***Sculpture:*** Sculpture is polymer glass restorative material. 1.0 nm diameter ceramic fillers are incorporated for achieving better strength. Polymerized with light and heat.

ii. ***Herculite:*** Herculite is microhybrid composite contain 0.6 nm ceramic fillers.

The composition and properties of composite inlay are tabulated in Table 19.6.

Cavity Preparation

The cavity preparation is by and large same as for gold inlays. Preparation for inlays, as usual, require creating a withdrawal form. The cavity design is a 'modified gold inlay preparation'.

The modifications are as follows:

1. 2.0 mm isthmus width at the occlusal aspect.
2. No less than 2.0 mm cavity depth at its most shallow point.
3. The axial wall depth should be minimum 1.0 mm (1.0–1.5 mm).
4. The primary flare angle (90–100°) of exit in the proximal walls (may not be in the embrasure areas).

Table 19.6: Indirect composite inlay system: Composition and properties

Brand name (manufacturer)	*Composition*	*Polymerization*	*Key points*
Art glass (Heraeus-Kulzer)	70 wt% filler of barium silicate glass (0.7 μm). 30 wt% organic resin matrix. Art glass contains four to six functional groups facilitating more double-bond conversions.	Photocuring is carried out in a special unit using a xenon stroboscopic light. The system emits 4.5 watts luminous power, initially for only 20 milliseconds, followed by rest of 80 milliseconds. The short excitation time followed by a longer period of non-exposure allows the already cured resin molecules to partially relax, allowing non-reactive double-bond carbon groups for reaction.	Used to fabricate inlay, onlays and crowns with/without metal substrate.

(*Contd...*)

Table 19.6: Indirect composite inlay system: Composition and properties (*Contd...*)

Brand name (manufacturer)	*Composition*	*Polymerization*	*Key points*
Belleglass HP and NG (Kerr Sybron Corporation) (Belleglass NG incorporates advanced nano-particles and submicron filler particles providing improved physical properties)	The base (dentin) composite has barium glass fillers (78.7% wt) and resin matrix; (BisGMA), whereas surface (enamel composite) has borosilicate fillers (74% wt) which provide enhanced optical characteristics. The resin matrix of enamel composite contains saturated methacrylate diurethane of TEGDMA and aliphatic dimethacrylate.	The base (dentin) composite is light cured, with a conventional light curing unit. The surface (enamel) composite is heat cured. The complete polymerization is carried by light curing followed by heating in an oven at 140°C at 80 psi for 20 minutes. The atmosphere is maintained oxygen free and under nitrogen gas pressure.	• The reduction in size of the filler improves the polishability and smoothness of the material. • Provides high flexure strength. • Excellent wear resistant • High translucency • Shrinkage approximately 0.9%
Sinfony (3M ESPE)	Polyfunctional methacrylate monomers, as resin matrix along-with Ultra-fine glass ceramic powder (strontium aluminium borosilicate glass—40% wt)—0.5 μm size as filler and as a microfiller. Pyrogenic silica as a microfiller.	Two polymerising units: Visio alpha is equipped with a halogen lamp and Visio beta is equipped with four fluorescent tubes. The polymerization wavelength ranges from 400 to 500 nm.The polymerization mode for alpha source is 40°C for 15 seconds; whereas, beta source is 40°C for 15 minutes.	Used for full veneers, inlays/onlays, glass fibre reinforced bridges and for the customization of prefabricated teeth. Pyrogenic silica, having large surface area provides thickening effect and also controls the rheological properties of the composite. Two sources of polymerization improves physical properties.
Targis-Vectris (Ivoclar Vivadent)	Resin matrix—conventional BisGMA and filler. Filler—77wt%, barium glass (1.0 μm). Along with spheroid silica (0.25 μm) and colloidal silica (0.05 μm).	Targis is coated with glycerine gel (Targis Gel) to prevent formation of oxygen-inhibited surface layer. Polymerized in 450–500 nm halogen light unit for 10 minutes followed by tempering in heat (95°C) for 25 minutes and cooling for 5 minutes.	The material can be used with and without metal framework to fabricate adhesive inlays/onlays/ veneers and crowns.
SR Adoro (Ivoclar Vivadent)	The components include SR Link (to bond to metal framework), a liner, dentin material, stains, incisal material and Opaquer. SR Link comprises a monomer that contains a highly hydrophobic hydrocarbon chain. The resin matrix consists of UDMA and TEGDMA along with fillers (63% by weight and 10–30 μm size). Upon polymerization, the copolymers get integrated into the	Polymerization is similar to Targis; Halogen light followed by heating.	SR Adoro is improved version of Targis. The phosphoric acid reacts with the metal or the metal oxide, forming a phosphate layer. The passivating phosphate layer on the metal surface soon becomes inert. The methacrylate group of the phosphoric acid reacts with the monomer components of S R Link, forms a copolymer, and provides a bond to the veneering resin.

(*Contd...*)

Table 19.6: Indirect composite inlay system: Composition and properties *(Contd...)*

Brand name (manufacturer)	*Composition*	*Polymerization*	*Key points*
	composite and a homogeneous composite with a high filler content is obtained.		
Solidex (Shofu)	Light cured composite. Resin matrix is co-polymer of multi-functional (25%) and conventional (22%) along with fillers (53%, 1.0 μm size silicone dioxide and aluminium - oxide and micro-ceramic fillers)	Polymerization is carried out by halogen lights, curing for 1–5 minutes at a temperature of 55°C.	Available in 16 shades, used for inlays/onlays, veneers, crowns and temporary prosthesis. Provides high abrasion resistance and is cost effective.
Sculpture Plus (Pentron)	Nano-hybrid composite for indirect restorations. Resin matrix is poly-carbonate dimethacry-late (PCDMA) along-with nano particle fillers such as borosilicate glass, alumina, zirconium silicate and initiators.	Polymerization is two way curing; 5 minutes of pressure curing followed by 3 minutes of curing under high intensity light.	Available in 16 shades, used for inlays/onlays, veneers crowns, etc. Esthetically much better.
Paradigm and Paradigm MZ 100 (3M ESPE)	Nano hybrid composite (ceramic blocks added to make Paradigm MZ 100) Resin matrix is cross-linked BisGMA and TEGDMA along with fillers (85% ultrafine zirconia-silica)	Conventional polymerization (cross-linking ensures thorough polymerization)	Available in 12 shades; wear resistance is superior (similar to enamel). Used for indirect restorations, core build-up, etc.
Vita Zeta (Vita Zahnfabrite)	Light curing composite. Resin matrix is UDMA, TEGDMA along with 44% fillers of silicone dioxide.	Polymerization is carried out with curing unit of wavelength 300–500 nm, at a temperature of 40°C.	Used for indirect restorations; has high abrasion resistance.
Pearleste EZ (Tokuyama–Tokyo)	Nanosized indirect composite. Resin matrix is combination of TEGDMA, UDMA and Bis-MPEPP (methylacryloxy polyethoxy phenyl propane) along with fillers (0.04–0.08 μm) zirconia-silica.	Polymerization is carried out by curing at high pressure mercury lamp (350–500 nm) for two minutes followed by heating for 15 minutes in an oven under pressure.	Nano-sized fillers provide high level of translucency and esthetics.

5. 8–10° occlusal taper exiting from the most gingival aspect to the occlusal cavosurface to allow passive seating of the restoration.
6. All internal line angles should be rounded.
7. Gingival margins should be a flat butt joint with no bevel.

Using fine diamond points, refine all margins to a smooth butt finish, particularly at the gingival floor and proximal walls. If the cusps require protection from occlusal load, they will need to be reduced in height by at least 2.0 mm at angle to the vertical, which is parallel to the original occlusal cuspal incline.

Table 19.7: Difference in cavity preparation

Gold	*Base metal*	*Composite*	*Ceramic*
Narrow outline	Narrow outline	Wider outline	Wider outline
Depth less	Depth less	Depth more	Depth more than composite
Short bevel at cavosurface (reverse bevel at gingival wall)	No bevel (even at gingival margin)	No bevel (even at gingival margin)	No bevel (even at gingival margin)
Taper (2–6°)	Taper (2–6°)	Taper (8–10°)	Taper (10–12°)
Definite line angles	Definite line angles	Rounded line angles	Rounded line angles
Primary flare at proximal box walls (90–100°); Margins in embrasures	Primary flare at proximal box walls (90–100°); Margins in embrasures	Primary flare at proximal box walls (90–100°); Margins may not be in embrasures	Primary flare at proximal box walls (90–100°); Margins may not be in embrasures

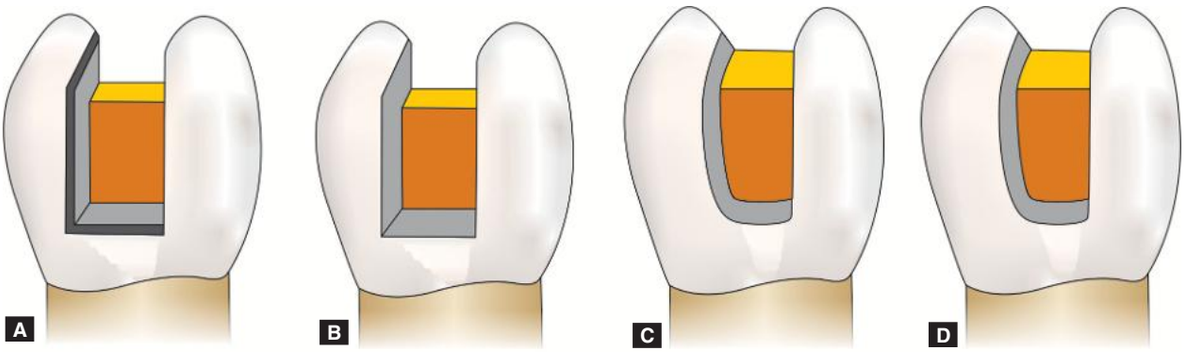


Fig. 19.43: Difference in cavity preparation: **(A)** Gold inlay; **(B)** Base metal; **(C)** Composite inlay; **(D)** Ceramic inlay

This will ensure an even thickness of composite over the cusp and avoid acute angles within the restoration, which may become stress points leading to initiation of fractures. Do not place bevels at the margins, because bevels will lead to fine fragile margins in the resin restoration.

The differences in cavity preparations for gold, base metals, composite and ceramic inlays are summarized in Table 19.7 and Fig. 19.43.

Method of Fabrication

The composite inlays can be fabricated either in office or in laboratory.

a. *In-office Fabrication*

The fabrication of composite inlay in office involves following steps:

- The prepared cavity is isolated preferably using rubber dams. The cavity walls are painted with a lubricant compatible with the hybrid composite material, which facilitates easy removal of inlay.
- A retainerless, contoured, clear matrix is adjusted around the proximal area of the prepared cavity. A clear reflecting wedge is placed at the interproximal gingival margins. The wedge is firmly placed to create rapid separation of the teeth. (Wedge compensates for thickness of the mylar matrix band facilitating developing contact area with the adjacent tooth.)
- The hybrid composite material is placed into the prepared cavity starting with the proximal box. The material is gently condensed with a ball burnisher. The occlusal portion of the preparation is also filled and gently condensed. The ball burnisher used for condensation should be coated with a resin adhesive to avoid sticking of the material.
- The pressure of the wedges helps rapid separation of the teeth and also do not allow excess material at the gingival area. The interproximal area is cured for 30 seconds both from facial and the lingual aspects; followed by curing of occlusal surface for 30 seconds (soft polymerization).
- After the completion of light curing, the inlay is gently eased out of the prepared cavity. (A gentle lift from the proximal area using hand instruments is usually required.) If the inlay resists removal, a loop of dental floss along with a small increment of the composite material is placed in the central fossa area of the inlay and light cured. The inlay is then pulled along the path of withdrawal.

- Fresh lubricant is painted on the inlay surfaces. The lubricant excludes air and allows the inlay to completely cure without an air-inhibited layer. The air-inhibited layer is the softest layer of composite resin and should be excluded. The inlay is light cured for an additional 60 seconds, followed by tempering in an oven at 110°C for 7 minutes. (The combined light and heat curing ensures complete polymerization of the material, provides increased hardness and wear-resistance. Polymerization shrinkage is also minimized.)

b. Laboratory Fabrication

The fabrication of composite resin inlay, carried out in the laboratory, involves the following steps:

- The rubber base material is used to make impression of the prepared cavity (Fig. 19.44A). A stone die is also prepared accordingly (Fig. 19.44B).
- The gingival contacts should remain intact during sectioning the cast even at the expense of the adjacent tooth. The impression can be poured again, if an additional cast is required.
- The preparation margins are outlined with a red pencil (Fig. 19.44C). A separating medium is applied to the internal surface of the die and also to the adjacent tooth. The separating medium is dried with a gentle air stream.
- The composite material of better viscosity and the correct shade is selected. Initially, the proximal area followed by occlusal area should be filled by gently condensing the material. The occlusal and proximal anatomy is carved simulating functional anatomy (Fig. 19.44D). Each surface is light cured for 60 seconds (Fig. 19.44E). The inlay is then removed from the die. The armamentarium is displayed in Fig. 19.44F.
- The inlay is cured in oven (Fig. 19.44G)
- The inlay is thoroughly cleaned ultrasonically in a water bath and dried. The characterizing stain is applied to pits and fissures with a brush and light cured for 40 seconds (Fig. 19.44H). Stain may also be applied when the composite resin is being layered into the preparation to develop internal characterization.
- The inlay is tried in the die for any irregularities, if any, and also the marginal fit. If the proximal surface needs to be adjusted, a 5.0 × 5.0 mm square piece of double-sided articulating film, held in an articulating forceps, placed against the proximal surface of the tooth that is next to the prepared

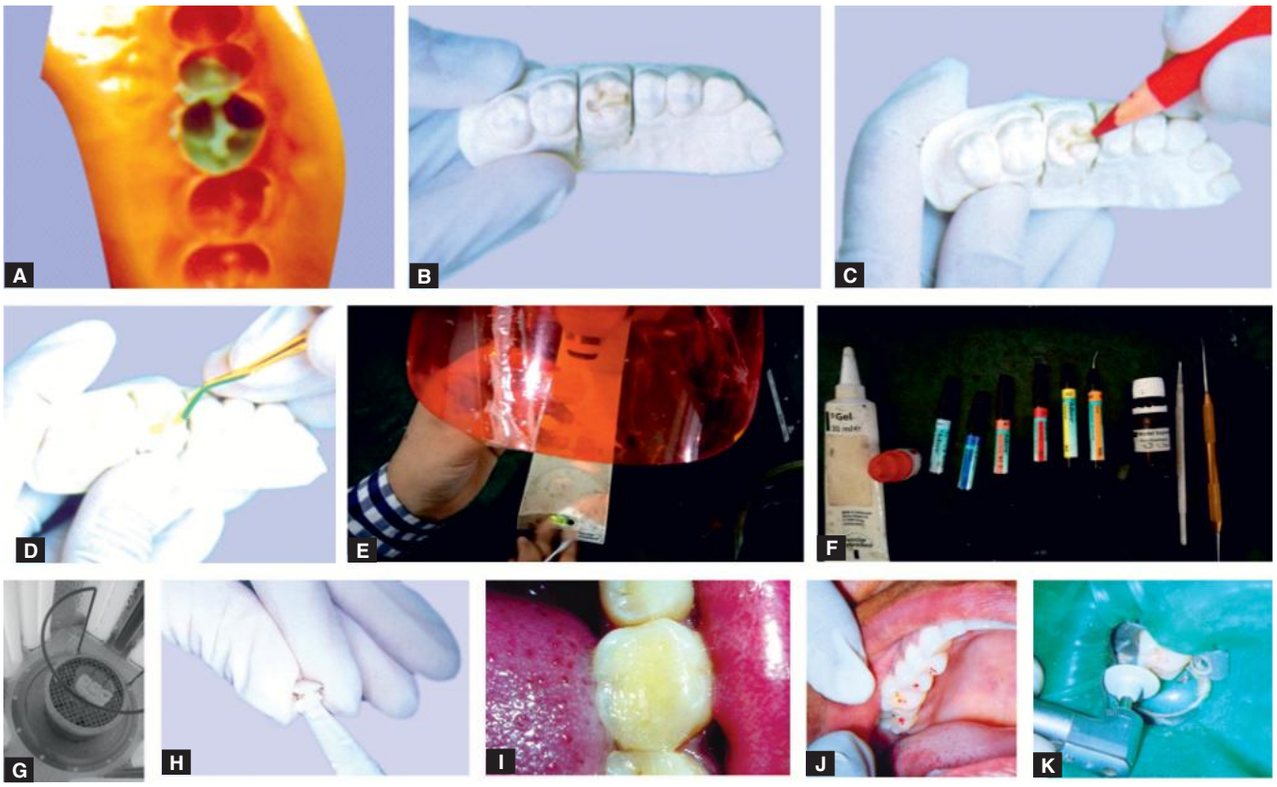


Fig. 19.44: Fabrication of composite inlay: **(A)** Impression of the cavity; **(B)** Die prepared from the impression; **(C)** Highlighting the margins; **(D)** Filling the cavity; **(E)** Curing of inlay; **(F)** Armamentarium used; **(G)** Preparing for oven curing; **(H)** Inlay cleaning and finishing; **(I)** Inlay tried in cavity; **(J)** Verifying occlusal relations; **(K)** Final finishing and polishing

tooth. The inlay is seated and then removed to evaluate the mark on the proximal surface. This area can be adjusted using Sof-Lex discs.

- The temporary filling material is removed from the cavity using hand instruments. The residual temporary filling material is further cleaned with a prophylaxis brush and slurry of pumice powder.
- The inlay is then placed in the prepared cavity and evaluated clinically (Fig. 19.44I). A pack of gauge must be placed around the teeth to protect the patient from aspirating the inlay.
- The tooth should be isolated before placing the inlay, preferably with rubber dam. It is advisable to isolate at least two teeth distal to the tooth being restored. This facilitates inlay seating since the clamp jaws might impinge on the gingiva or interproximal area of the tooth. A heavy-weight rubber dam should be used to afford optimal gingival retraction during cementation.
- After seating the inlay, verify that the margins of the inlay coincide with the cavosurface margins of the tooth preparation. The excess material is trimmed off using safe-sided abrasive points.
- The occlusal relationships both in centric occlusion and in lateral movements are verified (Fig. 19.44J). If occlusal adjustment is necessary, the surface can be re-characterized. The final polish of the restoration and refinement of the margins should be carried out after bonding.
- Each restoration needs to be evaluated for marginal integrity, proximal contact relationship with adjacent teeth, occlusal relationship with the opposing arch and color. The color of the cement should be compatible with the selected shade for the inlay.
- The inner surfaces of the inlay are roughened to allow for chemical crosslinking between the cementing agent and the composite.
- The cavity is thoroughly cleaned using slurry of pumice powder. The enamel margins are etched for 20 seconds, and washed with a gentle water spray. The dentin and etched enamel are pretreated with an aluminum oxalate dentin conditioner for 30 seconds. The dual-cure cement is mixed in accordance with the manufacturer's instructions and applied to the tooth preparation and also the internal surface of the inlay. Alternatively, the ultrasonics are used to place and bond the inlays. The ultrasonic technique takes advantage of the thixotropy of the composite resin. As soon as ultrasonic energy is applied, a highly viscous composite turns into a low viscous material. The clinicians can identify and remove the overhangs easily, because the excess composite gets as viscous as it was before as soon as the ultrasonic energy is discontinued. The restoration is light cured from all aspects; proximal, facial, lingual, and occlusal, for 60 seconds each. After polymerizing the cement, margins of the restoration are finished using multifluted finishing burs in sequence. The restoration is finished and polished with an aluminum oxide polishing disc. The final polish is accomplished with a composite resin polishing paste (Fig. 19.44K). The rubber dam is removed and the occlusion is verified with articulating paper in centric occlusion and excursive movements. If the occlusion is adjusted, then the restoration is re-polished.

REPAIR OF COMPOSITE RESTORATIONS

Repair implies partial replacement of the composite restoration allowing preservation of rest of the restoration, which presents no clinical or radiographic evidence of failures. The repair procedure will preserve the tooth structure and maintain function of the restoration.

Rationale

- Preservation of tooth structure
- Increased longevity of the restoration
- Reduction of potentially harmful effects on the dental pulp
- Reduction in treatment time
- Reduced cost to the patient
- Less risk of iatrogenic damage to the adjacent tooth

The possibility of obtaining a chemical bond between resin in the old substrate and the bonding agent decreases with time. The resin does not release the radicals for bonding. (Hydrolytic degradation is higher in old composites.) The inorganic filler particles of old composite need to be resilanized to improve bond to organic monomer of the repair material.

Various methods have been suggested to establish adequate bond strength between the existing composite and the new composite. These treatments include, surface roughening and the use of intermediate bonding agents to enhance repair bond strength. The roughening of the surface through acid etching, burs or abrasives, promotes mechanical interlocking through microretentive features and application of bonding agent improves surface wetting and chemical bonding with the new composite. In addition to surface roughening, silane treatment of the exposed filler particles can be carried out as silane acts as a bifunctional molecule that chemically binds to silica particles (through formation of siloxane bonds) of older composite and methacrylate groups of

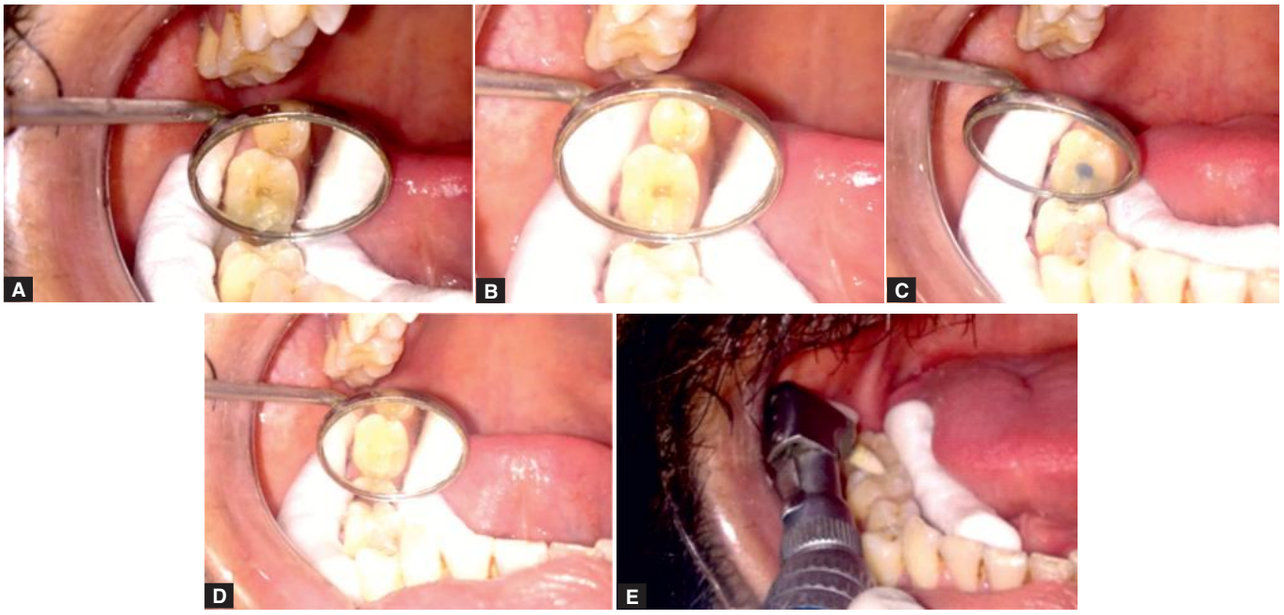


Fig. 19.45: Repair of composite inlay: **(A)** Preparation of the fractured area; **(B)** Prepared cavity; **(C)** Acid etching; **(D)** Prepared area restored; **(E)** Finishing of restoration

organosilane forms covalent bonds with resin monomer of newer composite.

Silanes have the ability to 'wet' inorganic material, essential for intimate contact for better bonding

The steps involved in repair process are:

1. *Surface treatment:* Various methods of surface treatments are:
   a. Abrasion
      i. Air abrasion with 50 µm aluminum oxide particles
      ii. Sandblasting with 30 µm silanated silica coated aluminum oxide particles
      iii. Abrasion with diamond bur
   b. Acid etching
      i. 37% phosphoric acid
2. *Clinical steps*
   - Remove the defective part of the composite restoration and any adjacent lesions of secondary caries.
   - Ensure adequate moisture control (rubber dam isolation)
   - Bevel the margins of the preparation, place 1.0 mm wide bevel on the margin of the composite resin to be repaired. (Bevels increase the surface areas for bonding and facilitate better esthetic outcome.)
   - Acid etch the composite resin substrate along with the adjacent tooth tissue with 37% phosphoric acid for 15–30 seconds. Wash thoroughly and dry the area.
   - Sandblast the composite resin substrate and exposed adjacent tooth tissue.
   - The silica-coated composite substrate surface is silanized by the application of the silane (allowed to dry for one minute).
   - An adhesive bonding system should be applied to the substrate and the adjacent tooth tissues.
   - Composite resin restorative material, compatible with the adhesive bonding system, is applied using an incremental technique to repair the defect with adequate photopolymerization.
   - Finish the repaired composite.
   - Final occlusal assessment of the restoration.

A case of repair of composite restoration is depicted in Fig. 19.45A to E.

BIBLIOGRAPHY

1. Abbott AP, Barron JC, Ryder KS, Wilson D. Eutectic-based ionic liquids with metal-containing anions and cations. Chem A Eur 2007;13:6495–501.
2. Alessandro DL, Reis A, Ballester RY. Polymerization shrinkage: effects of constraint and filling technique in composite restorations. Dent Mater 2004;20:236, 2004–40.
3. Aljabo A, Neel EAA, Knowles JC, Young AM. Development of dental composites with reactive fillers that promote precipitation of antibacterial-hydroxy-apatite layers. Mat Sci Engg 2016;60:285–92.
4. Alquidaihi FS, Cook NB, Diefenderer KE, Bottino MC and Platt JA.: Comparison of internal adaptation of Bulk-fill and increment-fill resin composite materials. Oper Dent 2019; 44, E32–E44.
5. Anand VS, Kavitha C, Subbarao CV. Effect of cavity design on the strength of direct posterior composite

restorations: An empirical and FEM analysis. Inter J Dent 2011;1–6.
6. Anjum A, Matin K, Uchida R, Tagami J. Influence of aging on direct resin composite repair bond. Int Chin J Dent 2008;8:35–40.
7. Bacchi A, Nelson M, Pfeifer CS. Characterization of methacrylate-based composites containing thio-urethane oligomers. Dent Mater 2016;32:233–39.
8. Baratieri LN, Ritter AV. To bevel or not in anterior composites. J Esthet and Restorat Dent 2005;17:264–67.
9. Beyth N, Farah S, Domb AJ, Weiss EI. Antibacterial dental resin composites. Reactive Function. Polymers 2013;1–8.
10. Borges ALS, Costa AKF, Saavedra GSFA, Komori PCP, Borges AB, Rode SM. Color stability of composites: Effect of immersion media. Acta Odontol Latinoam 2011; 24:193–99.
11. Boulden JE, Cramer NB, Schreck KM, Couch CLO, Troconis C, Stansbury JW, Bowman CN. Triolane-methacrylate composites as dental restorative materials. Dent Material 2011;27:267.
12. Bourbia M, Ma D, Cvitkovitch DG. Cariogenic bacteria degrade dental resin composites and adhesives. J Dent Res 2013;92:989–94.
13. Brueckner C, Schneider H, Haak R. Shear bond strength and tooth-composite interaction with self-adhering flowable composites. Oper Dent 2017;42:90–100.
14. Bruzell Roll EM, Jacobsen N, Hensten-Pettersen A. Health hazards associated with curing light in the dental clinic. Clin Oral Investig 2004;8:113–17.
15. Cardoso PC, deOliviera AR, Lopes LV, Cabral SC, Beatriz M. In vivo evaluation of different techniques for establishment of proximal contacts in posterior resin composite restorations. Braz J Oral Sci 2011;10:12–16.
16. Chatzistavrou X, Velamakanni S, DiRenzo K, Lefkelidou A, Fenno JC, Kasuga T, Boccaccini AR, Papagerakis P. Designing dental composites with bioactive and bactericidal properties. Mat Sci Engg 2015;52:267–72.
17. Chen MH. Update on dental nano-composites. J Dent Res 2010;89:549–54.
18. Cherchali FZ, Mouzali M, Tommasino JB, Decoret D, Attik N, Aboulleil H, Seux D, Grosgogeat B. Effectiveness of the DHMAI monomer in the development of an antibacterial dental composite. Dent Mater 2017;1–11.
19. Cramer NB, Stansbury JW, Bowman CN. Recent advances and developments in composite dental restorative materials. J Dent Res 2011;90:402–08.
20. da Veiga AM, Cunha AC, Ferreira DM. Longevity of direct and indirect resin composite restorations in permanent posterior teeth: a systematic review and meta-analysis. J Dent 2016;54:1–12.
21. Dafar MO, Grol MW, Canham PB, Dixon SJ, Rizkalla AS. Reinforcement of flowable dental composites with titanium dioxide nanotubes. Dent Mater 2016;32:817–26.
22. Darmani H, Al-Hiyasat AS, Milhem MM. Cytotoxicity of dental composites and their leached components. Quint Int 2007;38:789–95.
23. de Medeiros Dantas LC, da Silva-Neto JP, Dantas TS, Naes LZ, das Neves FD, da Mota AS. Bacterial adhesion and surface roughness for different clinical techniques for acrylic polymethyl methacrylate. Int J Dent 2016;1–6.
24. Demarco FF, Collares K, Coelho-de-Souza FH, Correa MB, Cenci MS, Moraes RR, Opdam NJM. Anterior composite restorations: A systematic review on long-term survival and reasons for failure. Dent Mater 2015; 31:1214–24.
25. Demirci M, Tuncer S, Sancakli HS, Tekce N, Baydemir C. Five-year clinical evaluation of a nanofilled and a nanohybrid composite in class IV cavities. Oper Dent 2018;43:261–71.
26. Dorfer CE, von Behlenfalvy ER, Staehle HJ, Pioch T. Factors influencing proximal dental contact strengths. Eur J Oral Sci 2000;108:368–77.
27. Doukoudakis S. Establishing approximal contacts in Class-II composite resin restorations. Oper Dent 1996; 21:182–86.
28. Duminis T, Shahid S, Karpukhina NG, Hill RG. Predicting refractive index of fluoride containing glasses for aesthetic dental restorations. Dent Mater 2018; 34:e83–e88.
29. Elshereksi NW, Ghazali M, Muchtar A, Azhari CH. Review of titanate coupling agents and their application for dental composite fabrication. Dent Mater J 2017;36: 539–52.
30. Ender A, Mehl A. Influence of scanning strategies on the accuracy of digital intraoral scanning systems. Int J Computer Dent 2013;16:11–21.
31. Fan C, Chu L, Rawls HR, Worling BK, Cardeon HL, Whang K. Development of an antimicrobial resin—a pilot study. Dent Material 2011;27:322–26.
32. Ferracane JL, Hilton TJ. Polymerization stress-is it clinically meaningful ? Dent Mater 2016;32:1–10.
33. Fonts AS, Di Mauro E, Dall'Antonio LH, Sano W. Study of the influence of pigments in the polymerization and mechanical performance of commercial dental composites. Rev Odontol Bras Central 2012;21:468–72.
34. Fronza BM, Ayres APA, Pacheco RR, Rueggeberg FA, Dias CTS, Giannini M. Characterization of inorganic filler content, mechanical properties, and light transmission of bulk-fill resin composites. Oper Dent 2017;42:445–55.
35. Fronza BM, Rueggeberg FA, Braga RR. Monomer conversion, microhardness, internal marginal adaptation, and shrinkage stress of bulk-fill resin composites. Dent Mater 2015;31:1542–51.
36. Gan JK, Yap AU, Cheong JW, Arista N, Tan CBK. Bulk-fill composites: Effectiveness of cure with poly- and monowave curing lights and modes. Oper Dent 2018; 43:136–43.
37. Garoushi SK, Lasilla LVJ, Vallittu PK. Fiber reinforced composite in clinical dentistry. Chinese Jr Dent Res 2009; 12:7–12.
38. Guo G, Fan Y, Zhang JF, Hagan JL, Xu X. Novel dental composites reinforced with zirconia-silica ceramic nanofibers. Dent Mater 2012;28:360–68.
39. Habib E, Wang R, Wang Y. Inorganic fillers for dental resin composites: present and future. ACS Biomater Sci Engg 2016;2:1–11.
40. Han SH, Park SH. Comparison of internal adaptation in class II bulk-fill composite restorations using Micro-CT. Oper Dent 2017;42:203–14.
41. Illie N, Rencz A, Hickel R. Investigations towards nano-hybrid resin-based composites. Clin. Oral Investig 2013; 17:185–93.
42. Imazato S. Antibacterial properties of resin composites as filled adhesives. Dent Mater 2002;18:227–31.
43. Jagger DC, Wilson NHF. Defective dental restorations: to repair or not to repair? Part I: Direct Composite Restorations. Dent Update 2011;38:78–83.

44. James A, Mackenzie L, Sanch P. The current status of materials for posterior composite restoration: the advent of low shrink. Dent Update 2009;36:401–07.
45. Jung JH, Park SH. Comparison of polymerization shrinkage, physical properties, and marginal adaptation of flowable and restorative bulk fill resin-based composites. Oper Dent 2017;42:375–86.
46. Kaizer MR, Almeida JR, Goncalves AP, Zhang Y, Cava SS, Moraes RR. Silica coating of non-silicate nanoparticles for resin-based composite materials. J Dent Res 2016; 95:1394–1400.
47. Khan AS, Azam MT, Khan M, Mian SA, Rehman I. An update on glass fiber dental restorative composites: A systematic review. Mater Sci Engg 2015;47:26–39.
48. Kopperud HM, Schmidt M, Kleven IS. Elution of susbtances from a silorane-based dental composite. Eur J Oral Sci 2010;118:100–02.
49. Landuyt KLV, Yoshiara K, Geebelen B, Peumans M, Godderis L, Hoet P, Meerbeek BV. Should we be concerned about composite nano-dust? Dent Mater 2012; 28:1162–68.
50. Leprince JG, Palin WM, Vanacker J. Physico-mechanical characteristics of commercially available bulk-fil composites. J Dent 2014; 42:993–1000.
51. Liebenberg WH. The axial bevel technique: A new technique for extensive posterior resin composite restorations. Quint Int 2000;31:231–33.
52. Lien W, Vandewalle KS. Physical properties of a new silorane-based restorative system. Dent Material 2010; 36:337–41.
53. Loomans BA, Opdam NJ, Roeters FJ, Bronkhorst EM, Plasschaert AJ. The long-term effect of a composite resin restoration on proximal contact tightness. J Dent 2007; 35:104–08.
54. Lowe RA. Advances in composite resin materials: The material science behind modern restoratives. Inside Dent 2015;2:1–6.
55. Lung CYK, Matinlinna JK. Aspects of silane coupling agents and surface conditioning in dentistry: an overview. Dent Mater 2012;28:467–71.
56. Maghaireh GA, Taha NA, Alzraikat H. The silorane-based resin composites: a review. Oper Dent 2017;42: e24–e34.
57. Magno MB, Nascimento GCR, da Roscha YSP: Silorane-based composite resin restorations are not better than conventional composites-a meta-analysis of clinical studies. J Adhes Dent 2016;18:375–86.
58. Mainjot A. Recent advances in compsite CAD/CAM blocks. Int J Esthet Dent 2016;11:275–80.
59. Marashdeh MQ, Gitalis R, Levesque C, Finer Y. Enterococcus faecalis hydrolyzes dental resin composites and adhesives. J Endod 2018;1–5.
60. Marcondes M, Souza N, Manfroi FB, Burnett LH, Spohr AM: Clinical evaluation of indirect composite resin restorations cemented with different resin cements. J Adhes Dent 2016;18:59–67.
61. Mikhail SS, Schricker SR, Azer SS, Brantley WA, Johnston WM. Optical characteristics of contemporary dental composite resin materials. J Dent 2013;41:771–78.
62. Milosevic M. Polymerization mechanics of dental composites-advantages and disadvantages. Procedia Engg 2016;149:313–20.
63. Miotti LL, Santos IS, Nicoloso GF, Pozzobon RT, Susin AH, Durand LB. The use of resin composite layering technique to mask discoloured background: A CIELAB/CIEDE2000 analysis. Oper Dent 2017;42:165–74.
64. Moszner N, Salz U. New developments of polymeric dental composites. Progress in Polymer Sci 2001;26:535–76.
65. Munhoz T, Fredholm Y, Rivory P, Balvay S, Hartmann D, da Silva P, Chenal JM. Effect of nanoclay addition on physical, chemical, optical and biological properties of experimental dental resin composites. Dent Mater 2016; 1–9.
66. Onisor I, Asmussen E, Krejci I. Temperature rise during photo-polymerization for onlay luting. Am J Dent 2011; 24:250–56.
67. Pfeifer CS. Polymer-based direct filling materials. Dent Clin N Am 2017;61:733–50.
68. Pitel ML. Low-shrink composite resins: a review of their history, strategies for managing shrinkage, and clinical significance. Compend. Contin Educ Dent 2013;34:578–90.
69. Price RBT. Light curing in dentistry. Dent Clin N Am 2017;61:751–78.
70. Rippe MP, Monaco C, Volpe L, Bottino MA, Scotti R, Valandro LF. Different methods for inlay production: Effect on internal and marginal adaptation, adjustment time, and contact point. Oper Dent 2017;42:436–44.
71. Rauber GB, Taguchi CMC, Padilha ACL,de Re Silveira RC, Bernardon JK and Barateiri LN: Color repair of a composite resin restoration. Oper Dent.: 2019; 44:1–7.
72. Rodrigues MC, Chiari MDS, Alania Y, Natale LC, Arana-Chavez VE, Meier MM, Fadel VS, Vichi FM, Hewer TLR, Braga RR. Ion-releasing dental restorative composites containing functionalized brushite nanoparticles for improved mechanical strength. Dent Mater 2018;34:746–55.
73. Romero MF, Haddock FJ, Freites AG, Brackett WW, Brackett MG. Restorative technique selection in class IV direct composite restorations: a simplified method. Oper Dent 2016;41:243–48.
74. Rueggeberg FA. State-of-the-art: dental photocuring-a review. Dent Mater 2011;27:39–52.
75. Safty S, Akhtar R, Silikas N, Watts DC. Nanomechanical properties of dental resin-composites. Dent Mater 2012; 28:1292–95.
76. Salehi S, Davis HB, Ferracane JL. Sol-gel-derived bioactive glass demonstrate antimicrobial effects on common oral bacteria. Am J Dent 2015;28:111–15.
77. Saunders SA. Current practicability of nanotechnology in dentistry Part I: Focus on nanocomposite restoratives and biomimetics. Clinical, Cosmetic and Investigational Dentistry 2009;1:47–51.
78. Shibasaki S, Takamizaa T, Nojiri K, Imai A, Tsujimoto A, Endo H, Suzuki S, Suda S, Barkmeier WW, Latta MA, Miyazaki M. Polymerization behaviour and mechanical properties of high-viscosity bulk fill and low shrinkage resin composites. Oper Dent 2017;42:e177–e187.
79. Shimokawa CAK, Sullivan B, Turbino ML, Soares CJ, Price RB. Influence of emission spectrum and irradiance on light curing of resin-based composites. Oper Dent 2017;42:537–47.
80. Shortall A, Palin W, Burtscher P. Refractive index mismatch and monomer reactivity influence composite curing depth J Dent Res 2008;87:84–88.
81. Silva PFD, Oliveira LRS, Braga SSL, Signori C, Armstrong SR, Soares CJ, Cenci MS, Faria-e-Silva AL. Effect of

selective carious tissue removal on biomechanical behaviour of class II bulk-fill dental composite restorations. Dent Mater 2018;1–10.
82. Soares CJ, Ferreira MS, Bicalho AA, de Paula Rodrigues M, Braga SSL, Versluis A. Effect of light activation of pulp-capping materials and resin composite on dentin deformation and the pulp temperature change. Oper Dent 2018;43:71–80.
83. Souza JCM, Bentes AC, Reis K, Gavinha S, Bucimeanu M, Henriques B, Silva FS, Gomes JR. Abrasive and sliding wear of resin composites for dental restorations. Tribiology Inter 2010;102:154–60.
84. Staxrud F, Dahl JE. Silanising agents promote resin-composite repair. Int Dent J 2015;65:311–15.
85. Tantbirojn D, Pfeifer CS, Braga RR. Do low-shrink composites reduce polymerization shrinkage effects? J Dent Res 2011;90:596–601.
86. Taubock TT, Marovic D, Zeljezic D, Steingruber AD, Attin T, Zrinka T. Genotoxic potential of dental bulk-fill resin composites. Dent Mater 2017;1–8.
87. Tomaszewska IM, Kearns JO, Ilie N. Bulk fill restoratives: to cap or not to cap-that is the question? J Dent 2015; 43:309–16.
88. Van Landuyt KL, Hellack B, Meerbeek BV, Peumans M, Hoet P, Wiemann M, Kuhlbusch TAJ, Asbach C. Nano-particle release from dental composites. Acta Biomater 2014;10:365–74.
89. Wang J, Dong X, Yu Q, Baker SN, Li H, Larm NE, Baker GA, Chen L, Tan J, Chen M. Incorporation of antibacterial agent derived deep eutectic solvent into an active dental composite. Dent Mater 2017;1–11.
90. Wang R, Habib E, Zhu XX. Evaluation of the filler packing structures in dental resin composites: From theory to practice. Dent Mater 2018;1–10.
91. Wang R, Habib E, Zhu XX. Synthesis of wrinked mesoporous silica and its reinforcing effect for dental resin composites. Dent Mater 2017;1–10.
92. Wang X, Huyang G, Palagummi SV, Liu X, Skritc D, Beauchamp C, Bowen R, Sun J. High performance dental resin composites with hydrolytically stable monomers. Dent Mater 2017;1–10.
93. Wang Y, Wang R, Habib E, Wang R, Zhang Q, Sun B, Zhu M. Surface modification of quartz fibres for dental composites through a sol-gel process. Mater Sci Engg 2017;74:21–26.
94. Weig KM, Filho TRM, Neto CAC, Costa MF. Evaluation of polymerization shrinkage of dental composites by an optical method. Mater Sci Engg 2015;47:70–76.
95. Xu HHK, Eichmiller FC, Antonucci JM, Schumacher LK, Ives LK. Dental resin composites containing ceramic whiskers and procured glass-ionomer particles. Dent Mater 2000;16:356–60.
96. Yancey EM, Lien W, Nuttall Cs, Brewster JA, Roberts HW and Vanderwalle KS.: Properties of nanofiber restorative composite. Oper Dent.: 2019; 44, 34–41.
97. Yu P, Yap AUJ, Wang XY. Degree of conversion and polymerization shrinkage of bulk-fill resin-based composites. Oper Dent 2017;42:82–89.
98. Zhang K, Zhang N, Weir MD, Reynolds MA, Bai Y, Xu HK. Bioactive dental composites and bonding agents having remineralizing and antibacterial characteristics. Dent Clin N Am 2017;61:669–87.
99. Zhang L, Weir MD, Hack G. Novel rechargeable calcium phosphate dental nanocomposite. Dent Mater 2016; 32:285–93.
100. Zhang N, Ma J, Melo MAS, Weir MD, Bai Y, Xu HK. Protein-repellent and antibacterial dental composite to inhibit biofilms and caries. J Dent 2015;43:225–34.
101. Zhang Q, Vigier KDO, Royer S, Jerome F. Deep eutectic solvents: syntheses, properties and applications. Chem Soc Rev 2012;41:108–46.
102. Zhao J, Milanova M, Warmoeskerken MMCG, Dutschk V. Surface modifications of TiO_2 nanoparticles with silane coupling agents. Colloids Surf A 2012;413:273–79.